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PALEOGEOGRAPHIC AND TECTONIC EVOLUTION OF SOUTHERN TIBET SINCE MIDDLE CRETACEOUS TIME: NEW PALEOMAGNETIC DATA AND SYNTHESIS

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Abstract. During the second French-Chinese paleomagnetic campaign in Tibet, in the summer of 1981, we sampled the middle Cretaceous red beds of the Takena formation and the Cenozoic volcanics of the Lingzizong formation. A detailed analysis of about 400 specimens of red beds reveals a stable component of normal magnetization N in the north of the Lhasa block (north of the Nyainqentanglha range) and two stable components in the south (LT, HT) where the red beds are more strongly folded and are unconformably overlain by the Lingzizong volcanics. All three components are carried by hematite. Blocking temperatures range from 400° to 600° C for LT and up to the Néel temperature of hematite for both N and HT. LT is shown to postdate part of the folding of the Takena, whereas N and HT both indicate a positive fold test at the 99% probability level. In addition, a small (less than 10% of the natural remanent magnetization) antiparallel component is superimposed in most samples and may be consistent with a model of long-term acquisition of a chemical remanent magnetization for the red beds. Analysis of the volcanics reveals a single stable component with mixed polarities, carried either by hematite or magnetite. The paleomagnetic directions indicate that (1) the Lhasa block stood at 12.5°±3° N by Upper Cretaceous and at 13.5°±6.5° N in the Paleocene-Eocene, thus confirming the limited displacement of the margin of Eurasia between Albian-Aptian (deposition of the red beds) and Eocene (onset of the India/Asia collision). These values imply crustal shortening by as much as 1900 ± 850 km north of the block (since the Upper Cretaceous) and by 1400 ± 1200 km south of the block (since the onset of the collision), (2) Southern Tibet formed a single tectonic unit by middle Cretaceous time, and (3) differential rotations are observed within the block and are related to the collision. These rotations are discussed in the frame of a recent propagating extrusion tectonics model. This study confirms the close paleogeographic relationship between Indochina (Khorat plateau) and Southern Tibet prior to the India/Asia collision.

1. Introduction

The second French-Chinese scientific expedition for the study of the structure and evolution of the northern side of the India-Asia collision zone was carried out during the summer of 1981 [Allègre et al., 1984]. The first campaign in 1980 had focused on the northern limit of the Indian plate along the Zangbo River (referred to as the Yarlung-Zangbo suture zone; Figure 1) and on the Gangdise mountains immediately to the north [Tapponnier et al., 1981a, b; Nicolas et al., 1981; Pozzi et al., 1982; Maluski et al., 1982; Westphal et al., 1983]. The goals of the 1981 campaign were both to solve some of the problems raised by the analysis of data provided by the previous campaign and to further our understanding of the structure of the Tibetan plateau by extending field studies to the north.

North of the Yarlung-Zangbo suture zone, the Tibetan plateau appears to be a composite terrane built of at least three different blocks [Chang and Cheng, 1973] (Figure 1). These independant tectonic units (or microcontinents) have presumably been accreted to the active southern margin of Eurasia during the Mesozoic [Chang and Cheng, 1973]. The mosaic pattern which results is reminiscent of the Western Cordillera of North America [Coney et al., 1980], although the scale and boundary conditions are quite different [Allègre et al., 1984]. According to Tapponnier et al. [1982] these blocks achieved their present position with respect to each other during the Cenozoic through a process of eastward "propagating extrusion" associated with major left-lateral strike-slip faulting. These movements are seen as the response of the southern margin of Eurasia to the collision of the Indian continent [Tapponnier et al., 1982, and unpublished manuscript,1984; Peltzer et al., 1982] and have been confirmed to some extent by paleomagnetic data in Southeast Asia [Achache et al., 1983a].

The 1981 French-Chinese campaign explored the southernmost Tibetan unit, referred to as the Lhasa block or southern Tibet. The block was explored from its southern boundary, the Yarlung-Zangbo suture, to its northern boundary, the Bangong-Nujiang suture (near Anduo, Figure 2). The occurrence of ophiolitic bodies within the Lhasa block near lake Nam Co (or Namuhu) raised the possibility of yet another suture within the Lhasa block [Tapponnier et al., 1981a; Marcoux et al., 1982; Girardeau et al., 1984]. New geochemical and radiometric age determinations provide constraints on the timing of activity in these sutures [Allègre et al., 1984].

This paper presents the paleomagnetic study of middle Cretaceous red beds and of lower Cenozoic

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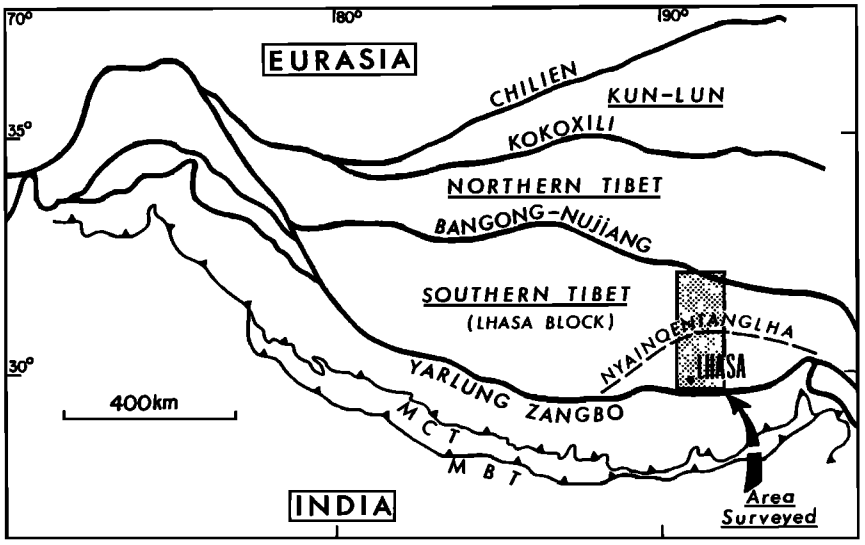


Fig. 1. Schematic tectonic map of the Tibetan plateau and surrounding area showing several independant tectonic blocks or microcontinents (name underlined), separated by suture zones (heavy lines). Barbed lines indicate continental underthrusting within the Indian continent. Also indicated is the Nyainqentanglha fault zone (dashed lined). Shading marks the area surveyed during the 1981 French-Chinese collaboration program (see details in Figure 2) [after Burg et al., 1984].

volcanics collected along the full latitudinal extent of the Lhasa block in 1981. Preliminary results have been reported by Courtillot et al. [1982], Achache et al. [1983b], and Allègre et al. [1984]. The new paleomagnetic data strongly constrain the geodynamical evolution of the Lhasa block in the last 100 Ma, that is most of its history since it was accreted to the Eurasian southern margin.

2. Geology of the Lhasa Block

For a detailed description of the geology (stratigraphy, tectonics) of the Tibetan plateau, the reader is referred to more specialized papers [Tapponnier et al., 1981a; Burg, 1983]. We shall only summarize their results here and discuss the elements which are relevant to our study.

The sedimentary cover of southern Tibet consists mainly of upper Paleozoic (Carboniferous and Permian) and Mesozoic deposits. It is intruded by a major east-west trending plutonic belt, the Gangdise batholith (consisting mainly of tonalites and granodiorites), and more localized granite bodies near Lhasa and Yangbajain. Ages for these intrusives range from 100 to 40 Ma [Allègre et al., 1984; Schärer et al., 1984]. South of the Gangdise belt and north of the Zangbo suture, we find a strongly folded and faulted series of Mesozoic rocks: the Xigaze series (Figure 2). The internal structure and continuity between this series and the sedimentary cover of the ophiolites to the south (radiolarites) are still debated [Allègre et al., 1984].

North of the granitic belt, the upper part of the Mesozoic deposits consists essentially of red sandstones and siltstones of the Takena formation, intercalated at their base with Orlitolina Limestone and calc-alkaline volcanics. The tropical facies of this formation becomes clearly more continental upward, with increasing occurrence of fossil plant remains (silicified woods), and suggests a delta environment. Despite their widespread occurrence on the Tibetan plateau, these red beds seem to be essentially confined to the Albian-Aptian period (100-110 Ma) [Jaeger et al., 1982].

North of Yangbajain (see Figure 2), the Lhasa block seems to be devoid of Cenozoic deposits, and the gently folded Takena formation is the youngest formation observed. In contrast, south of Yangbajain, Mesozoic formations including the Takena are strongly folded and unconformably overlain by thick series of younger volcanics, essentially andesites and ignimbrites of the Lingzizong formation. This angular unconformity is enhanced by the occurrence at the base of the Lingzizong of red conglomerates intercalated with volcanic tuffs. An $^{39}Ar/^{40}Ar$ geochronological study on andesites near Linzhu gave an age of 60 Ma [Maluski et al., 1982], and a K/Ar study on ignimbrites sampled between Lhasa and Yangbajain led to an age of 48 Ma [Westphal et al., 1983]. The contrasting features between the northern and southern parts of the Lhasa block (intensity of folding of the Mesozoic deposits and localized occurrence of Cenozoic volcanics) suggest a rather different tectonic history for the two parts of the block. Recent fieldwork confirms a rather mild tectonic activity north-west of the Nyainqentanglha range (Figure 2), strongly contrasting with the observations made southeast of this range [Marcoux et al., 1982; Tapponnier et al., 1984]. This together with microtectonic observations of intense deformation (penetrative shearing and rotational deformation [Mattauer et al., 1982]) suggest the occurrence of large relative displacements within the Lhasa block along a major strike-slip fault associated with the Nyainqentanglha [Tapponnier et al., 1984].

3. Previous Paleomagnetic Work

Previous paleomagnetic studies in the Lhasa block were mostly carried out on the Cretaceous

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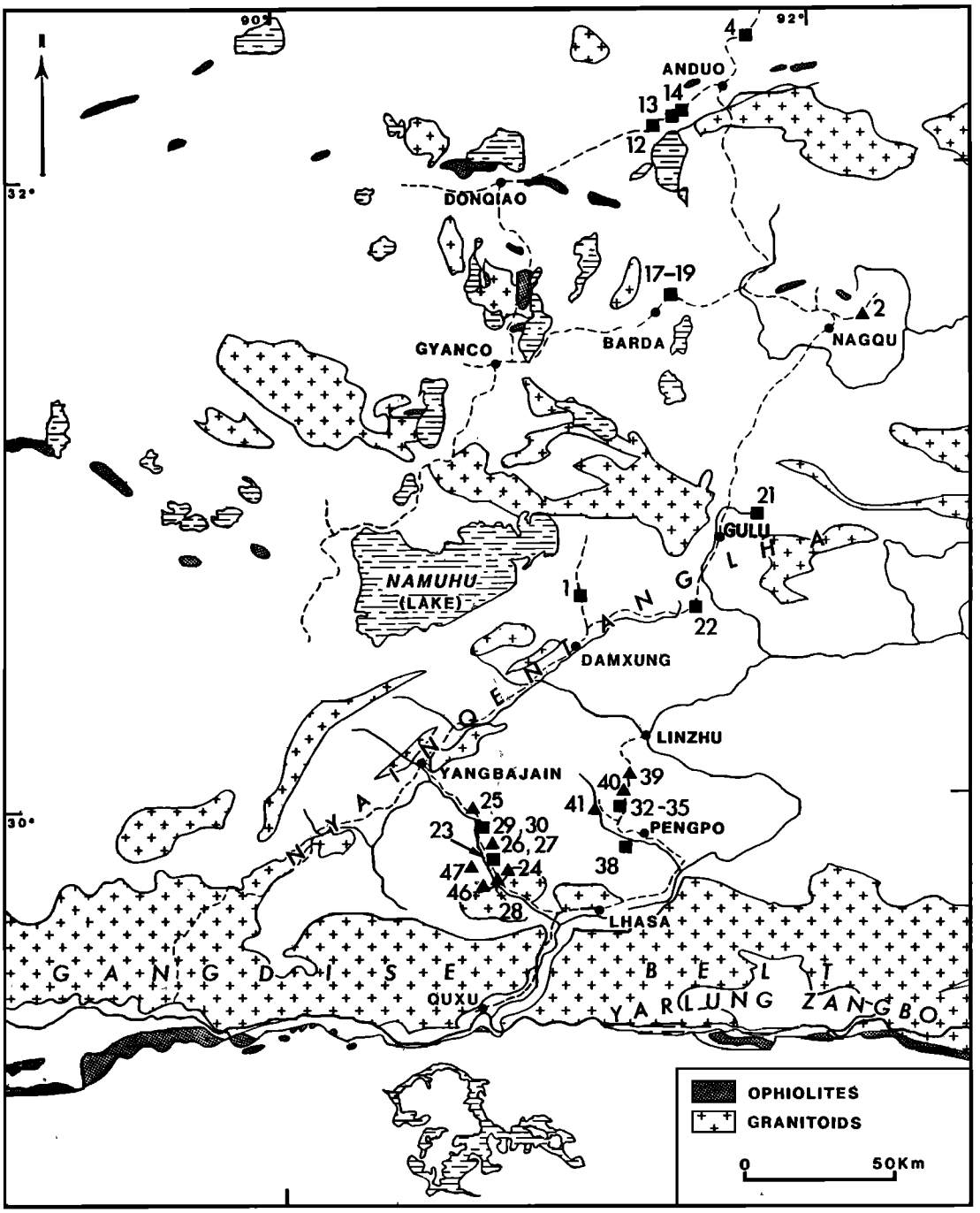


Fig. 2. Simplified geologic map of the area surveyed during the 1981 campaign, showing the major ophiolite and granite outcrops. Paleomagnetic sampling sites are indicated (squares, Takena red beds; triangles, volcanics). The Nyainqentanglha fault zone, also indicated, separates two zones with contrasting geology (see text).

red beds from the Takena formation [Zhu et al., 1977, 1981; Pozzi et al., 1982; Wesphal et al., 1983]. Studies from Zhu et al. [1977, 1981] cannot be considered as reliable since the magnetic directions were obtained using alternating field demagnetizations only, which prove to be unfit to decipher the magnetization of red sandstones (see Zhu et al., [1981, Figure 3] and this paper). Pozzi and his colleagues [Pozzi et al., 1982; Westphal et al., 1983] sampled the Takena formation in the area of Lhasa and Linzhu (Figure 2), where it is most strongly folded, interbedded with volcanics, and overlain by the younger Lingzizong volcanics. Striking similarities between the apparent polar wander path (APWP) of India and the virtual geomagnetic poles (VGP) of these red beds, before unfolding, led Pozzi et al. [1982] to suggest the possibility of successive remagnetizations of the Takena formation. These authors, however, finally concluded from a positive fold test that remagnetization, if present, must have been only partial. In the same papers two sites of Cenozoic volcanics from the Lingzizong formation were

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studied and led to two different directions of the paleofield. These directions will be carefully discussed below and compared to the new results derived in this paper.

In addition, a single preliminary study has been published on Jurassic rocks [Zhu et al., 1981], which suggests an equatorial position of Southern Tibet at that time.

4. Sampling

Sampling was performed using an electrically powered portable drill with standard 1-inch-diameter drill bits. In some instances, the sampling of a site was extended by the collection of oriented blocks. When the difficulties of access to a site prevented us from using the drill, a complete collection of blocks was taken. Between five and sixteen cores and/or blocks were taken in each site leading to a total of 240 independent samples (150 red beds, 90 volcanics). The cores were oriented using both a magnetic and a solar compass, whereas only a magnetic compass was available for orienting the blocks. For all sites the mean difference between the measured magnetic declination and the computed solar declination was less than 2° (with standard deviation less than 2°) except for site 40 (mean of -2.3 and standard deviation (s.d.) of 3.0) and site 21 (mean of 1.5 and s.d. of 2.4). Up to four specimens were taken from each core or block leading to a total of about 600 specimens.

Takena

We have considerably extended the sampling of the Takena formation. The area now sampled covers almost the entire latitudinal extent of Southern Tibet essentially along a 300-km-long meridian cross section at the longitude of Lhasa (figure 2). The distribution of sites should allow us to determine the extent of possible remagnetizations and to study possible relative displacements within Southern Tibet. Such displacements could be associated either with younger (Cenozoic) suturing within the block (as suggested by the presence of ophiolites near Lake Nam Co) or with the activation of large transcurrent faults (e.g., the Nyainqentanglha). Extensive resampling in the south should allow us to determine possible links between the more active tectonism and volcanism and the paleomagnetic record of the red beds.

Altogether, eighteen sites of Takena red beds were sampled, ten north of Yangbajain and eight in the south near Lhasa or Linzhu. Sites 1 and 22 are located on the trace of the Nyainqentanglha fault. Site 4 is situated to the north of the Bangong-Nujiang suture zone and, therefore, does not belong to the Lhasa block, as defined above.

Lingzizong

This formation has only been observed in the area of Lhasa and Linzhu where it has been sampled at nine different sites. In addition, one site was taken in the Lhasa granites (site 46). A single site of andesites and tuffs (site 2) has been sampled in the north, near the town of Nagqu (Figure 2) but has not been dated yet, leading to a total of eleven sites of volcanics or intrusives.

5. Magnetic study

Methodology

Measurements of remanent magnetization on the 600 specimens are performed using a digital Schönstedt spinner or a CTF cryogenic magnetometer. Progressive destruction of the magnetization is achieved either by heating in a furnace built in our laboratory or by using a Schönstedt single-axis alternating field demagnetizer.

As a first step in the analysis of the magnetization in each site, up to three pairs of specimens, each pair being taken from the same core, are submitted to careful progressive demagnetization. One specimen from each pair is submitted to successive steps of heating and cooling in a field free space up to 700° C, while the other is submitted to alternating fields (af) with increasing peak values up to 100 mT, along three orthogonal axes. This pilot study allows us to assess the main characteristics of the magnetization.

On the basis the results of this study a routine demagnetization procedure is then derived. This procedure is designed to minimize the number of demagnetization steps necessary to isolate the stable components of the remanence.

The magnetic components are first identified by visual inspection of the demagnetization curves, using standard orthogonal projections [e.g. Zijderveld, 1967]. A mean direction is then computed in the range of blocking temperatures where the component has been identified. To do so, we use a modified version of Kirschvink's [1980] principal component analysis which allows us to fix this range.

Takena Formation

NRM. Intensities of natural remanent magnetization are rather well grouped, spanning only 1 order of magnitude from 2×10^{-3} to 4×10^{-2} A/m except for sites 1, 4, and 30, which show intensities as low as 5×10^{-4} A/m. Different specimens taken from the same core or block generally display consistent intensities of magnetization. Furthermore, no correlation is observed between the depth to surface of each specimen and its intensity within the same core, suggesting that the small variations in intensity which are observed are due to mineralogic heterogeneities rather than to surface weathering.

The analysis of NRM directions already provides some information on the magnetization of these rocks. Indeed, NRM directions are well grouped within most sites and even provide a preliminary fold test, as shown in Figure 3.

Pilot study. The pilot study confirmed the first observations made on the NRM and revealed the main features of the magnetic record of the Takena formation. First, it revealed a rather different general behavior of sites from the northern and southern parts of the Lhasa block, as could be anticipated from geological observations (see above). According to this

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analysis the samples have been separated into several groups.

Group I: This larger group includes most sites from the north (12, 13, 14, 17, 18, 19, 21, 22) and some from the Lhasa-Linzhu area (23, 32, 33, 38), that is, altogether, 12 out of the 18 sites sampled in the Takena formation. In all test specimens from these sites, af demagnetization up to 85-mT peak value removes less than 10% of the total magnetization and does not indicate a stable direction. On the other hand, thermal treatment consistently reveals a stable remanence with widely distributed blocking temperatures between 200° C and the Néel temperature of hematite ~680° C) (Figure 4). Multicomponent magnetization seems to be present in most test specimens, but the directions involved are probably close to each other and therefore lead to a rather monotonous demagnetization path except for site 12, where an intermediate polarity component seems to remain at very high temperatures (Figure 4a). A small random component with low blocking temperatures (below 400° C) is observed in some specimens but appears not to have any global significance. It can also be observed on Figure 4 that between 20 and 50% of the remanence is still present after heating at 600° C and that above this temperature a single component of magnetization seems to be preserved. This final component, however, is in some instances slightly offset from the origin (see sample 18-07A in Figure 4a), suggesting a complex behavior of the magnetization at very high temperatures. The progressive build-up of a random viscous remanent magnetization (VRM) at high temperatures (between 660° and 700° C) with intensities well above the noise level of the magnetometer prevents us from conducting a more detailed analysis of the last stages of demagnetization (see sample 23-03A in Figure 4b). Such a VRM is likely to be associated with mineralogic transformation (oxydation?) at high temperature, producing magnetic minerals in the superparamagnetic grain size range. It appears, therefore, that in order to isolate all the components with high blocking temperatures a careful progressive demagnetization has to be performed on all specimens above 600° C.

Group II: This group includes all remaining sites from the southern part of the Lhasa block (29, 30, 34, 35). The specimens analyzed reveal a clear multicomponent magnetization. One component is partly removed by af treatment but more than 50% of the NRM is preserved at 70 mT (Figure 5b). In sites 29 and 30 the intensity of magnetization after treatment at 100 mT is even larger than that of the NRM, indicating the superposition of two components with opposite directions of magnetization, one of them only being affected by the treatment (Figure 5a). Thermal demagnetization of the companion specimen confirms the existence of at least two components (with opposite polarities in sites 29 and 30 (Figure 5a)). Again, the final direction does not always point toward the origin; the magnetic viscosity of these samples seems to be higher than that of the previous group, leading to a rather unstable demagnetization path.

Group III: This very small group contains only two sites: site 1, located on the trace of the Nyainqentanglha fault, and site 4, which probably does not belong to the Lhasa block. The af demagnetization of the test specimens from these two sites reveals incoherent multicomponent magnetizations with final destructing field above 80 mT. Similar components are generally observed during thermal demagnetization, although, again, this procedure leads to rather noisy curves.

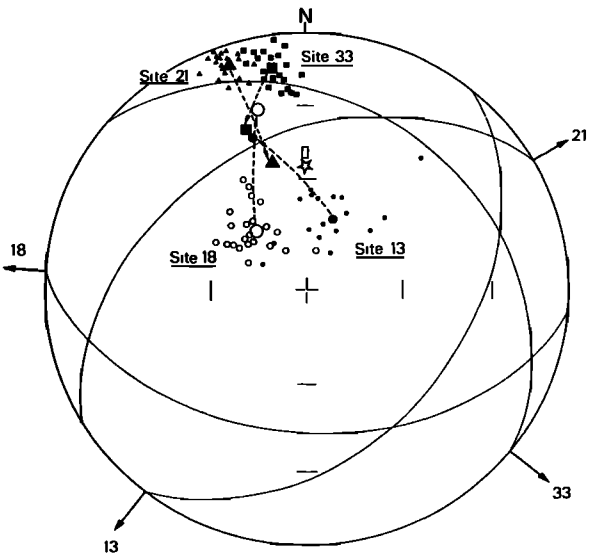


Fig. 3. Equal-area projection of NRM directions of individual specimens from four different sites located hundreds of kilometers apart in the red beds of the Takena formation (see Figure 2 for location). Great circles represent the intersection of bedding planes with the upper hemisphere. These NRM directions already indicate a positive fold test. Open square, present field direction; star, present dipole field direction.

Antiparallel superimposed magnetizations. Further demagnetization experiments were necessary in order to complete the pilot study. Four specimens from site 13 were submitted to progressive thermal demagnetization above 600° C (by 10° steps). These experiments revealed a very peculiar but systematic feature in the demagnetization curves of the specimens: between, say, 550° and 650° C, very little of the total remanence is removed (Figure 6). In addition, several components with rather different directions seem to be superimposed in this range of blocking temperatures, leading to a very erratic path of demagnetization. The same type of behavior is observed in other sites, although sometimes in a slightly different range of temperatures (Figure 7). Demagnetization curves display a range of behavior going from full "reversal" (13-03A, Figure 6, and 14-02B, Figure 7) to simple inflexion toward either shallower (18-08B, Figure 7) or steeper (13-04A and 19-07A, Figures 6 and 7) inclination, with westward (19-07A, Figure 7) or eastward (13-04A, Figure 6) declination. All curves can be explained by a single mechanism, namely, the occurrence of a single superimposed reversed (nearly antiparallel) component showing some scatter in direction and some variability in the range of blocking temperatures. An alternate explanation would involve several (at least two) distinct components with normal polarities and would require a large variability in the direction of

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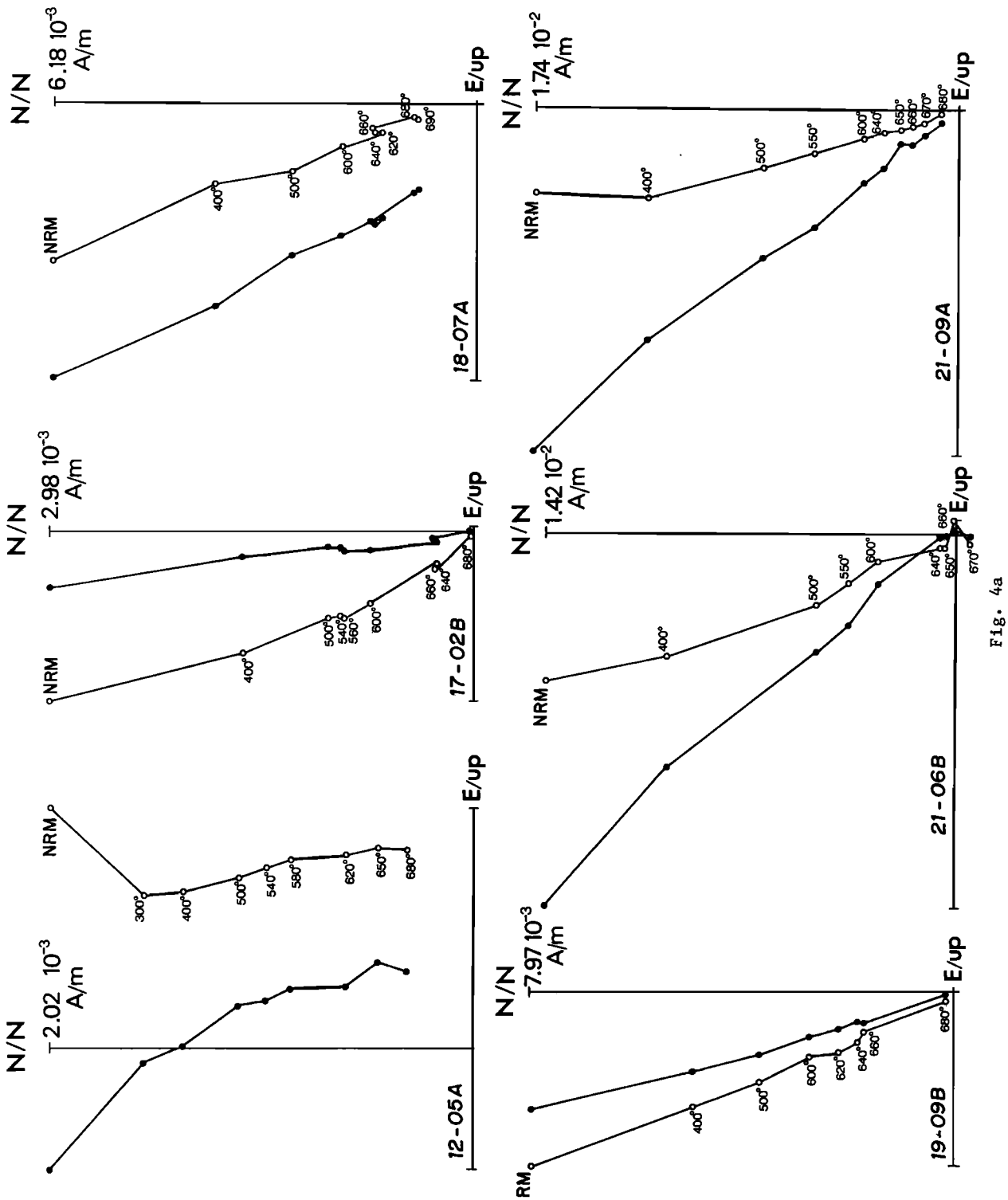


Fig. 4a

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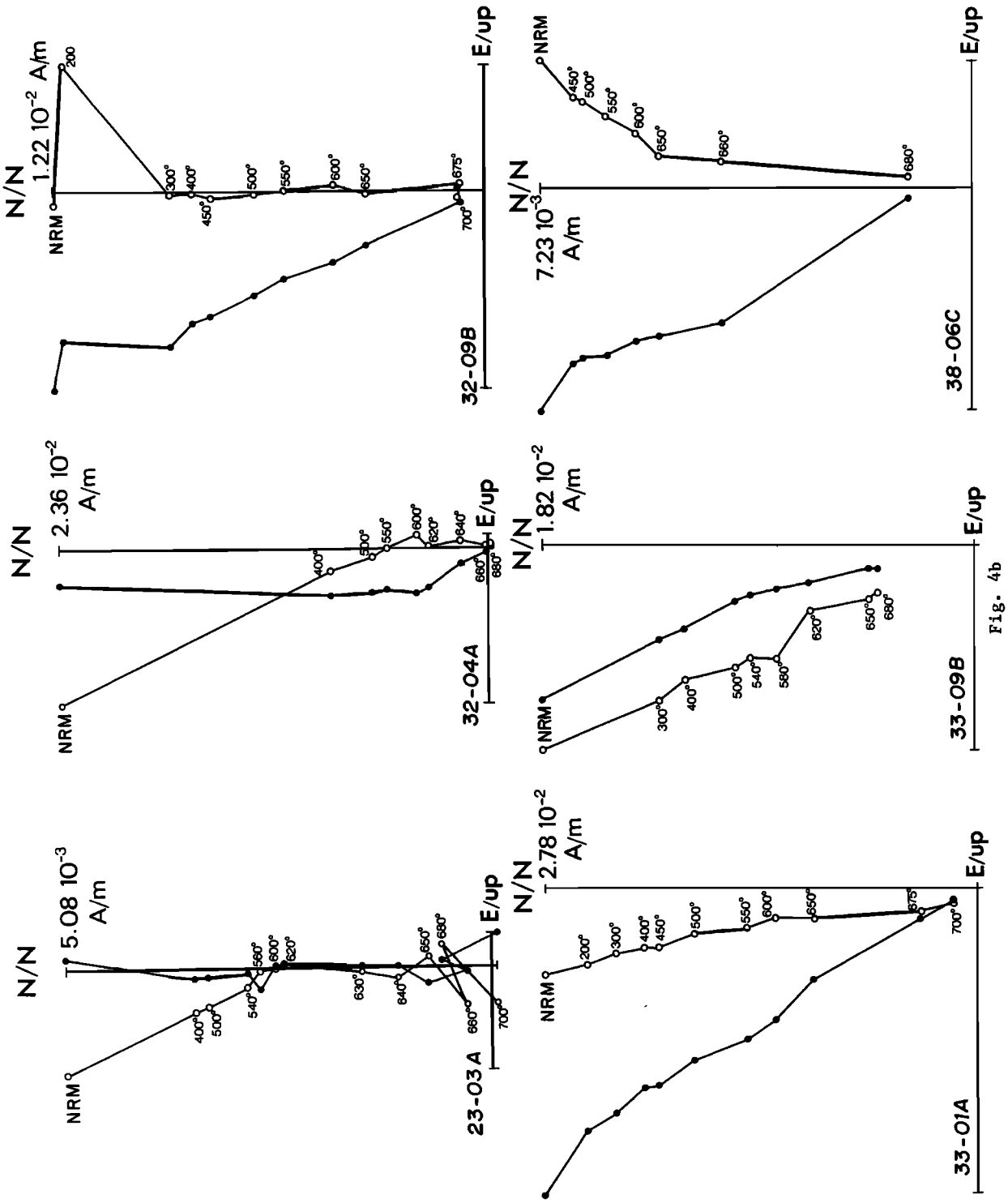


Fig. 4b

Fig. 4. Orthogonal vector projection of thermal demagnetization of red bed specimens from group I (see text). (a) Northern part of the Lhasa block. (b) Southern part of the Lhasa block. Solid symbols, projection on the north-south vertical plane. Open symbols, projection on the horizontal plane. Treatment is always indicated near the horizontal projection (in degrees Celsius).

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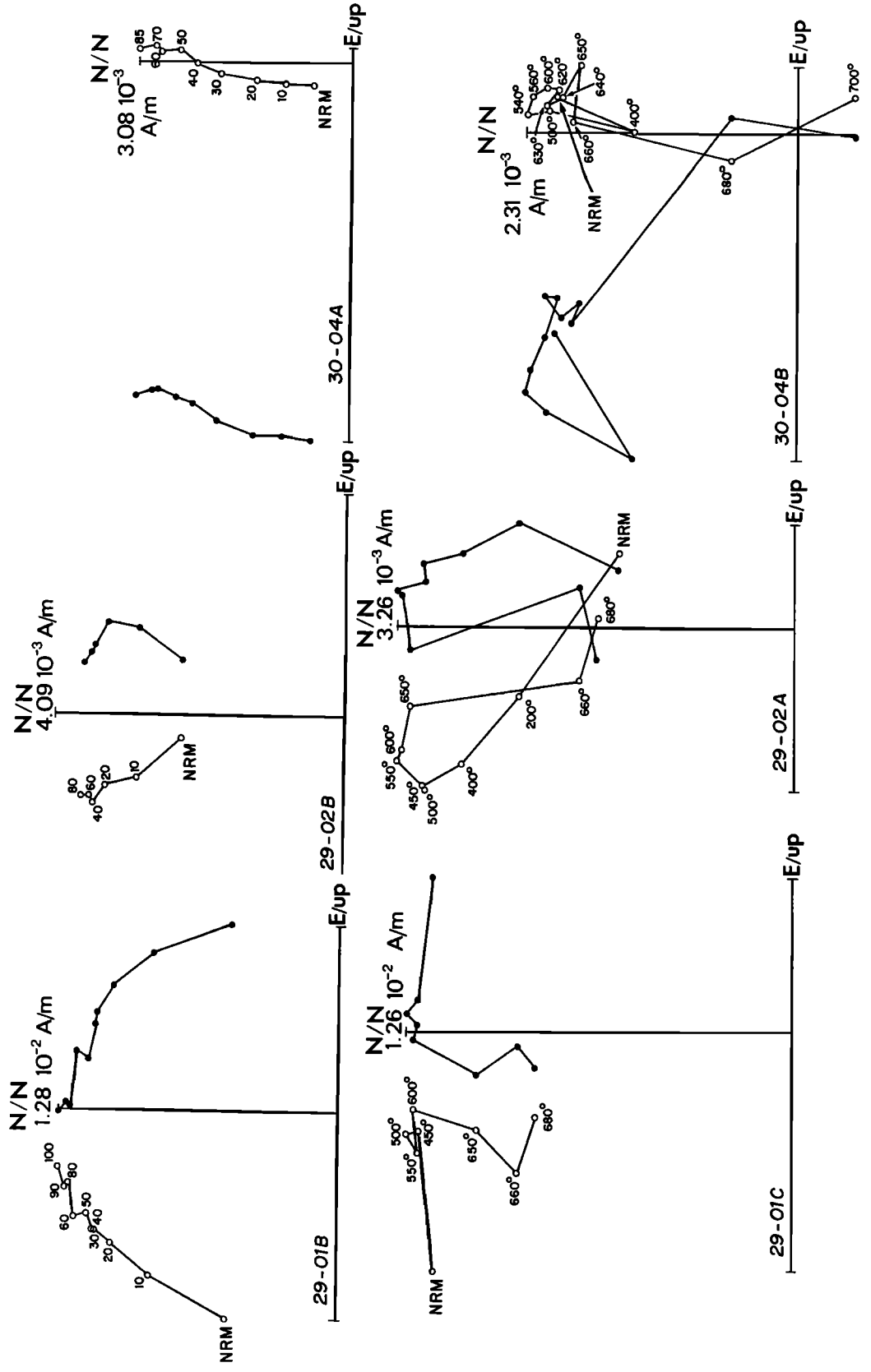


Fig. 5a

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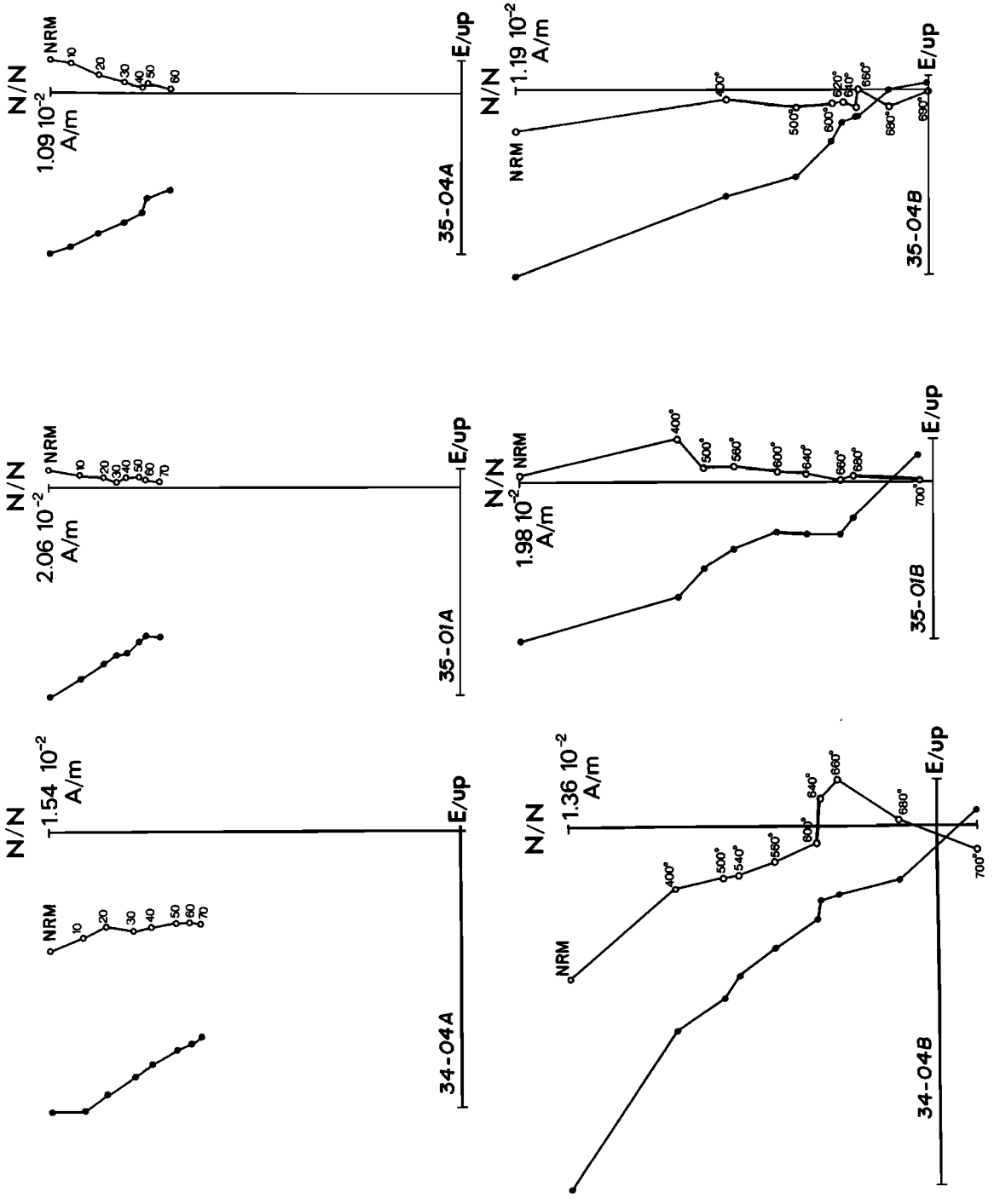


Fig. 5b

Fig. 5. Orthogonal vector projection of af (above) and thermal (below) demagnetization of red bed specimens from group II (see text). (a) Lhasa Yangbajain area. (b) Linzhu-Peng Po area. Treatment is in milliteslas for af diagrams and in degrees Celsius for thermal demagnetizations (see Figure 4 for conventions).

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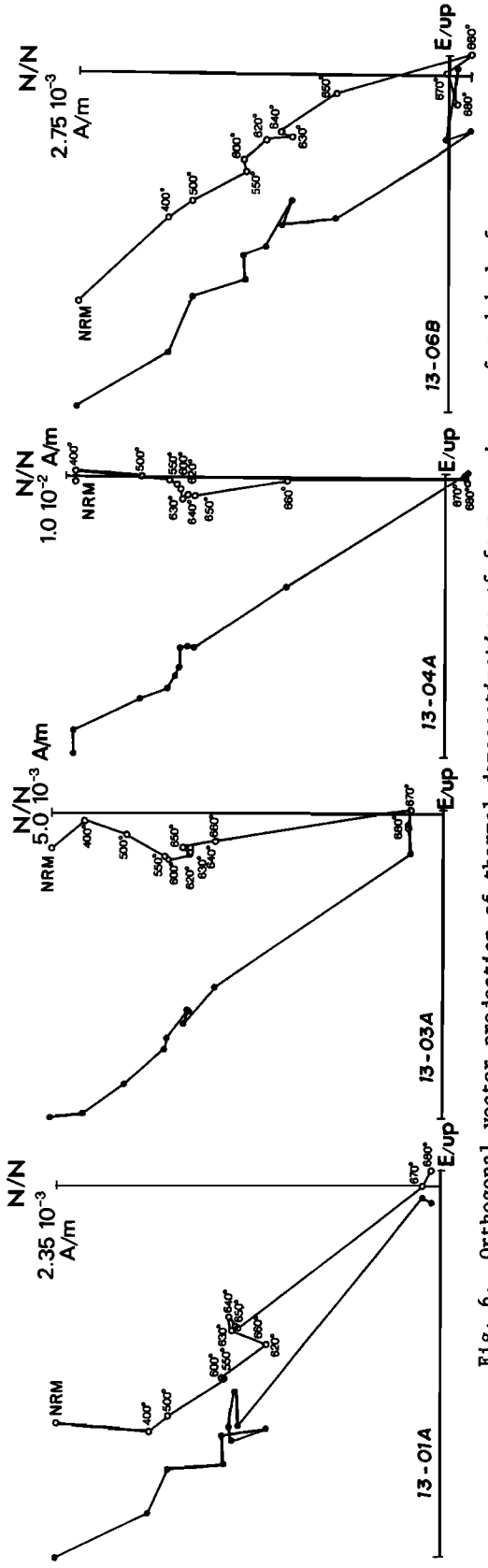


Fig. 6. Orthogonal vector projection of thermal demagnetization of four specimens of red beds from site 13, showing various effects of a small nearly antiparallel component with a narrow spectrum of blocking temperatures (see Figure 4 for conventions).

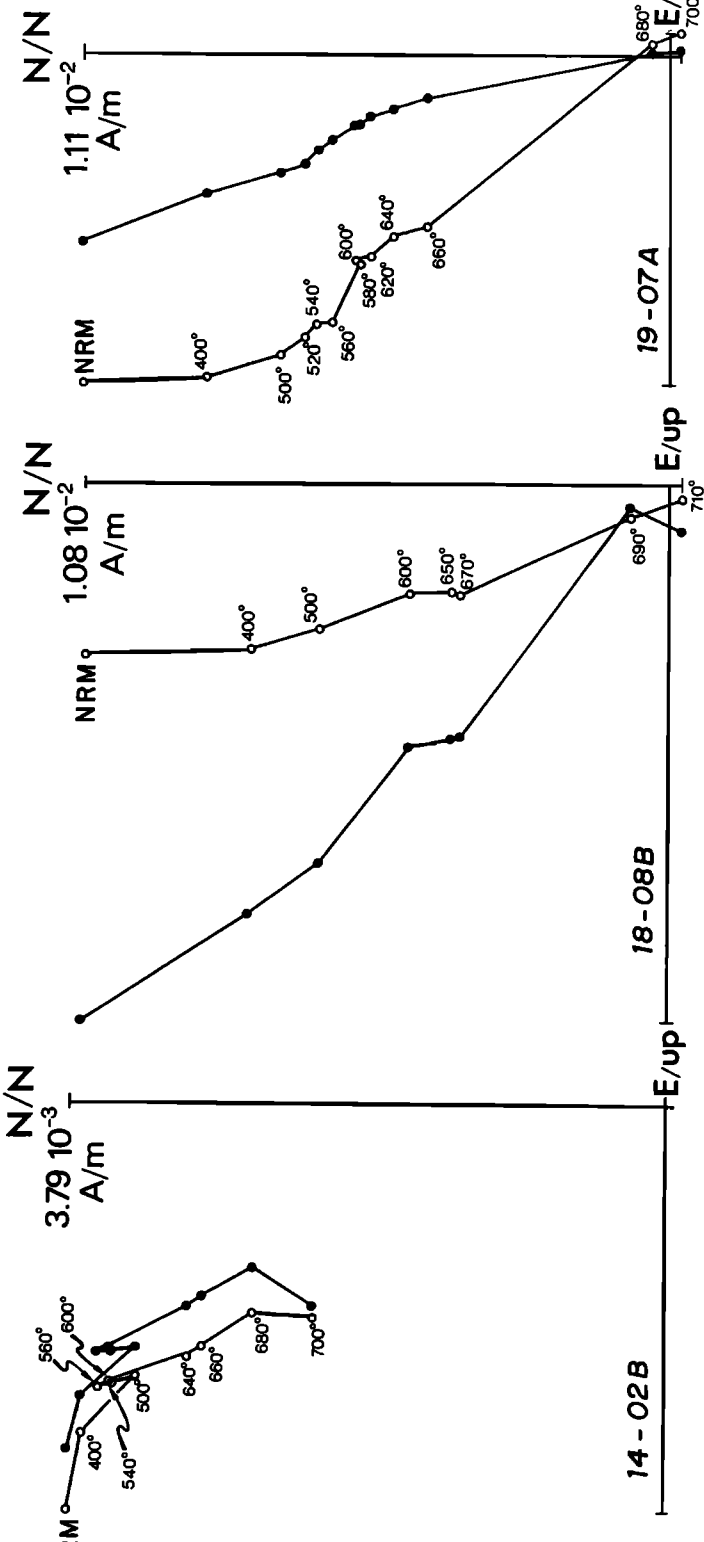


Fig. 7. Same as Figure 6 in three different sites (see Figure 4 for conventions).

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TABLE 1. Takena Formation : Northern Region

Site	N	Geographic Coordinates D,deg	Geographic Coordinates I,deg	Stratigraphic Coordinates D,deg	Stratigraphic Coordinates I,deg	α_{95},deg	K	VGP λ,°N	VGP ϕ,°E
12	8	6.3	51.7	-28.1	36.0	11.8	22.9	62.1	342.4
13	10	5.4	57.5	-20.0	30.1	5.9	68.6	65.8	324.7
14	5	-8.4	30.7	-13.2	26.5	11.5	45.3	68.2	308.1
17	5	-40.5	41.5	-28.5	9.1	5.8	173.5	52.0	322.1
18	10	-46.9	63.8	-23.0	27.9	7.7	40.6	63.1	328.2
19	6	-39.8	48.1	-25.7	21.6	8.7	60.0	58.7	326.5
21	13	-20.3	-0.6	-16.1	36.8	3.9	114.1	72.2	329.9
Mean	7	-20.2 (a)	43.9	-22.2	27.0	8.1	56.3	63.4	326.2 (b)
Mean w.o. 12	6	-24.1 (c)	41.9	-21.3	25.4	8.9	57.0	63.5	325.4 (d)
22	5	91.8	-4.3	90.8	-7.4	19.7		16.0	

N,number of samples (or sites); D,declination; I,inclination; A_{95},α_{95},radius of the circle of 95% confidence; K,precision parameter; λ,ϕ,geographic coordinates of the geomagnetic pole (VGP).
(a) α_{95}=20.2°,K= 9.9 .
(b) A_{95}= 5.9°,K=105 .
(c) α_{95}=23.3°,K= 9.2 .
(d) A_{95}= 6.5°,K=106 .

one of these components, even within a single site. Small heterogeneities in magnetic mineralogy can account for variations in the range of blocking temperatures of a given component as well as a small scatter in direction but not for large changes in this direction. Furthermore, a single reversed magnetization is quite simply explained from a geomagnetic point of view, whereas several distinct normal directions would lead to overcomplicated geodynamic or geomagnetic schemes.

Repeated observations of the kink illustrated in Figures 6 and 7 show that this reversed component is randomly distributed about a direction opposite to the dominant normal component and thus should not affect the determination of the mean direction of this normal component.

Results. In the first section of this paper, we recalled that important geological differences exist between the southern and northern parts of the Lhasa block. In order to trace possible consequences of this contrasting history on the magnetization of the Takena formation the results from the two regions have been analyzed separately.

Naturally, we have tried to eliminate the elusive reversed component discussed above. When this component is characterized by a narrow spectrum of blocking temperatures and therefore observed in a localized interval of the demagnetization process, one can stay away from this component and easily isolate the remaining stable components of magnetization. In some instances, however, the spectrum of the reversed component (or, rather, its overlap with the spectrum of the other component) may be quite large and an averaging procedure is provided by the principal component analysis, as described in the methodology.

Northern region. All sites from the north, apart from sites 1 and 4, belong to group I. In this group, with the only exception of some samples from site 12, a single stable component of magnetization (N) is consistently observed above, say, 400°C. When the superimposed reversed component is not (or hardly) observed, the mean direction is computed between 500°C and the origin (or 680°C when the direction misses the origin). In some cases, however, component N can only be isolated in the final part of the demagnetization process, sometimes above 600°C (see site 13, Figure 6). The mean of the directions obtained is then calculated for each site (Table 1) and plotted in Figure 8, before and after tectonic corrections are applied.

The within-site precision parameter for site 12 decreases from 54 to 23 when bedding corrections are applied. The mean direction for this site prior to bedding correction is not significantly different from the present field direction. This makes site 12 suspect of

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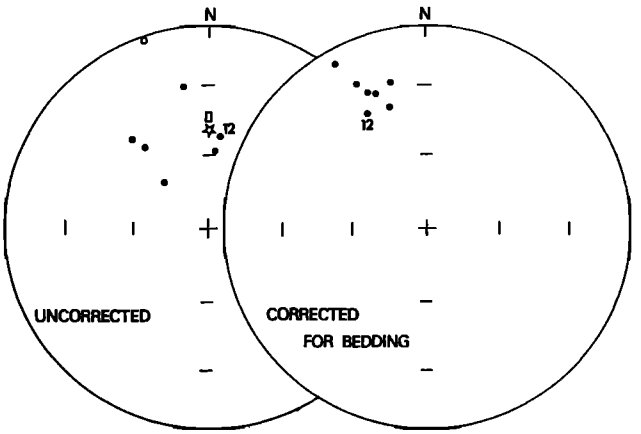


Fig. 8. Equal-area projection of mean directions of sites in the northern part of the Lhasa block before and after correcting for bedding (direction N). Open square, present field direction; star, present dipole field direction.

remagnetization, a proposal which agrees with the coarse lithology of the site. In order to avoid potential interference of a remagnetized direction the mean direction for northern sites should exclude site 12. This mean direction successfully passes the fold test at the 99% probability level [McElhinny, 1964]. We find D = 339°, I = 25° (α_{95} = 9°, K = 57). However, inclusion of site 12 happens not to change these results significantly: the overall mean is within 2° of the previous one and the statistical parameters and fold test are about the same (Table 1).

Routine demagnetization of samples from sites 1 and 4 confirmed the conclusions of the pilot study: many samples did not reveal any stable component of magnetization, and all were subject to viscous behavior, preventing us from determining a stable mean direction of magnetization.

As for site 22, an intermediate direction is observed with poor statistical parameters, both before and after tectonic corrections(Table 1; α_{95}=20°). This could result from superposition of components with opposite polarities and comparable intensities and would be consistent with the suggestion that this site may well be younger than other red bed sites in Southern Tibet (J.J. Jaeger, personal communication, 1981), thus most likely to have acquired a reversed component after the Cretaceous long normal interval. Alternatively, since site 22 is located on the Nyainqentanglha fault zone, it may have undergone a significant local rotation associated with strike-slip motion on the fault.

Southern region. In this area, both types of magnetizations from group I (sites 23, 32, 33, 38) (Figure 4b) and group II (sites 29, 30, 34, 35) (Figure 5) are observed. All samples show two distinct components of magnetization: first, a low blocking temperatures component (between 400° and 600° C) which will be referred to as LT, followed by a high blocking temperature component (above 620° C) referred to as HT (Figures 4b and 5). The antiparallel component does not have a marked effect in those sites.

The site mean directions for the HT component, both before and after tectonic corrections are applied, are given in Table 2 and Figure 9 and an overall mean is then computed: this fold test is positive at the 99% probability level (K increases from 2.3 to 46, for eight sites), implying that the HT component of magnetization (D = 354°, I = 23°, α_{95} = 8.3°, K = 46) was acquired prior to the folding of the Takena formation.

The LT component is revealed by thermal

TABLE 2. Takena Formation : Southern Region (HT Component)

Site	N	Geographic Coordinates D,deg	Geographic Coordinates I,deg	Stratigraphic Coordinates D,deg	Stratigraphic Coordinates I,deg	α_{95},deg	K	VGP λ,°N	VGP φ,°E
23	12	33.2	52.4	-0.8	15.1	8.4	27.4	67.9	272.9
29	9	-18.1	-56.0	-21.7	11.4	9.3	31.9	58.4	315.3
30	7	170.9	53.1	13.5	25.5	10.1	36.8	69.3	230.8
32	9	16.9	64.3	1.1	27.0	6.5	64.4	74.3	267.4
33	9	-11.8	6.4	-19.1	24.4	7.7	45.9	65.4	321.0
34	5	1.9	7.6	-7.5	29.0	14.0	31.0	73.9	298.3
35	4	-0.8	41.4	-6.0	24.1	14.3	42.4	71.7	290.3
38	6	-5.8	-28.6	-4.2	21.7	11.5	34.6	70.9	284.0
Mean	8	1.4 (a)	25.1	-5.7	22.6	8.3	45.7	71.2 (b)	288.4

See Table 1 for symbols.
(a) α_{95}=48.3°, K= 2.3 .
(b) A_{95}= 7.9°, K=50 .

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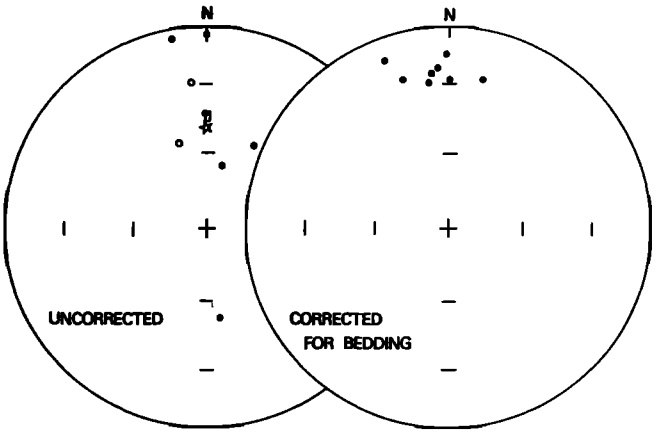


Fig. 9. Equal-area projection of mean HT components of magnetization of the red beds in the southern part of the Lhasa block before and after correcting for bedding. Open square, present field direction; star, present dipole field direction.

demagnetization only in sites 23, 32, 33, 38 (group I, Figure 4b) and both by thermal and A F treatment in sites 29, 30, 34, 35 (group II, figure 5). In six sites (23, 32, 33, 34, 35, 38) this LT component is characterized by a normal polarity, whereas in sites 29 and 30 it is clearly reversed (Figure 5a). These two sites, which are located about 500 m apart on the two flanks of a syncline, are sampled in one of the most strongly folded outcrops of the Takena (in site 30 the bedding is vertical and even locally overturned). Again, the mean directions are calculated in each site for the LT component in geographic and stratigraphic coordinates (Table 3, and Figure 10). The overall site mean is then computed, including sites 29 and 30 (antidirections are taken for these two sites). When tectonic corrections are applied, the statistical parameter K increases from 3.8 to 8.5 (ratio: 2.23) and the direction obtained is D = 357°, I = 22°, α_{95} = 20°. According to McElhinny's [1964] criterion this fold test is not significant at the 95% probability level (the ratio of precision parameters should be at least equal to 2.48). In other words, the dispersion of directions has been reduced by unfolding the formation but not sufficiently to conclude that the magnetization was acquired prior to folding with a confidence level of 95%. This can be simply interpreted by assuming that the component was acquired after part of the folding of the Takena had been completed, as will be discussed in more detail in the next section. It further implies that the LT component is younger than the HT component.

Lingzizong Formation

NRM. As opposed to the red beds, intensities of magnetization of the volcanic rocks span several orders of magnitude. Andesites (sites 2, 24, 26, 27, 39, 41, 47) show intensities between 10^{-2} and 10 A/m. Ignimbrites tend to have smaller intensities of magnetization ranging from 5×10^{-4} to 3×10^{-2} A/m (sites 25, 28), except in site 40 which shows very consistent values of NRM

TABLE 3. Takena Formation : Southern Region (LT Component)

Site	N	Geographic Coordinates D,deg	Geographic Coordinates I,deg	Stratigraphic Coordinates D,deg	Stratigraphic Coordinates I,deg	α_{95},deg	K
23	13	3.7	62.3	-19.1	10.6	7.9	28.4
29	8	163.3	25.5	164.8	-40.6	21.3	7.7
30	9	207.3	-67.2	193.1	27.0	14.3	14.0
32	9	-6.4	60.8	-9.4	21.4	10.8	23.4
33	10	-10.8	11.2	-20.3	29.1	5.7	71.5
34	5	0.0	22.1	-18.7	40.3	9.2	70.0
35	3	22.6	45.8	14.9	31.7	12.8	93.4
38	4	29.6	-37.0	28.7	14.9	18.6	25.2
Mean	8	4.0 (a)	28.5	-2.9	21.8	20.1	8.5
Mean of normal sites	6	6.2 (b)	30.2	-3.8	25.9	18.7	13.8

See table 1 for symbols.
(a) α_{95}=33.0°, K=3.8 .
(b) α_{95}=36.1°, K=4.4 .

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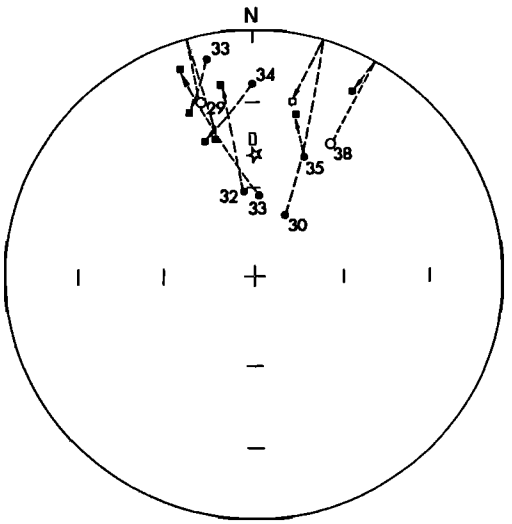


Fig. 10. Effect of the tectonic corrections on the BT component of magnetization of the red beds in the south of the Lhasa block showing an imperfect fold test (overcorrections and undercorrections). Open square, present field direction; star, present dipole field direction.

intensities (from 4×10^{-1} to 7×10^{-1} A/m). The granite samples have intensities of magnetization between 3×10^{-2} and 2×10^{-1} A/m.

Five sites (25, 26, 27, 28, 40) show an excellent grouping of NRM directions. The mean direction of sites 25 and 27 is close to the present field direction before any tectonic correction is applied. Site 40 points to a normal direction of magnetization (declination close to zero and downward inclination), whereas sites 26 and 28 point to a reversed one (except for core 28-01 which is normal), both before and after tectonic corrections are applied. Three out of the four flows of site 2 also show rather well-grouped directions of NRM close to the present field before corrections are applied, the directions from the lower flow being scattered. Finally, sites 24, 39, 41, 46, and 47 display generally more scattered directions, mixed polarities (antiparallel) being suggested in some of them (24, 41, and 46).

Pilot study. Due to the great heterogeneity in petrology (andesites, ignimbrites, granites) and in geographic position (and probably in age) of the volcanic rocks sampled, a large variety of behaviors were observed during demagnetization. Even within one single site, the pilot study was sometimes inconclusive and led us to carry out a systematic analysis of two specimens per core, one by af and the other by thermal treatments.

Despite this large variety of behaviors, the volcanics may be classified into three different groups.

Group A (sites 2, 24, and 27; Figure 11): The samples from this group consistently lead to a stable demagnetization curve when treated in alternating fields up to 80 mT (sometimes 100 mT), although such a treatment is not sufficient, in some cases, to remove entirely the original remanence (see 02-13B, 24-01B, 24-04B, Figure 11).

Progressive thermal demagnetization, in turn, sometimes leads to very erratic paths (see 24-01A, 24-04A, Figure 11). This behavior clearly seems to be linked with the presence of a secondary component. The carriers of the secondary (less stable) component may undergo mineralogic transformations during heating, perhaps giving birth to new magnetic carriers in the superparamagnetic range. This would allow the buildup of a viscous component induced by the presence of a small residual field in the cooling space or more likely during transportation from the furnace to the magnetometer. Although such is not always the case and stable single-component thermal demagnetization curves are also observed (see 02-13A, Figure 11), a routine af demagnetization is preferred for these sites.

Group B (sites 26, 41, 47, and 40; Figures 12 and 13): The preliminary analysis of these samples shows that af demagnetization, although still revealing part of the stable magnetization, is sometimes unable to remove more than 20% of the original remanence (see 26-06B, Figure 12).

Thermal demagnetization consistently reveals several components. The more stable one is at least in part carried by hematite with blocking temperatures above 600° C (Figures 12 and 13). Only one sample (41-07B, Figure 13) is fully demagnetized at 550° C, with magnetite as the sole carrier of magnetization. This sample comes from a flow which stands at a crest top in the vicinity of which large deviations of the magnetic compass were observed. However, the magnetic direction from site 41 is very stable both for thermal and af demagnetizations (Figure 13).

Strictly speaking, site 40 should not be included in group B since af as well as thermal demagnetization proves efficient in revealing entirely the different magnetic components (see 40-01A,B; 40-02A,B; Figure 13). It should be noted, however, that the higher blocking temperature component ($550 < T_b < 690$° C) is found to have the lower coercivities ($20 < B_c < 40$mT) (B_c is the corresponding destructing field),whereas the lower blocking temperature component ($300 < T_b < 500$° C) corresponds to higher coercivities ($B_c > 60$ mT). Despite its range of blocking temperatures, the former component may not be of primary origin and may be associated with later chemical alteration of the ignimbrite. Indeed, observation of thin sections of this volcanic body reveals a high degree of alteration and recrystallisation, and absolute age determination could not be obtained from this formation (C. Coulon, personal communication, 1983). Accordingly, all these sites, including site 40, have been routinely demagnetized by heating up to the Néel temperature of hematite.

Group C (sites 25, 28, 39, and 46; Figure 14): The samples from this third group display more variety than the previous ones and could be further divided into two smaller groups. Their remanent magnetization is essentially removed by thermal demagnetization below 600° C and thus thought to be due to magnetite. Similarly, af treatment removes more than 80% of this remanence in fields of 80 mT peak value or less. However, whereas in sites 28 and 39 both treatments seem to be equally efficient in revealing the components of demagnetization (see 28-04A,B and 39-04B,C; Figure 14), neither treatment appears to be adequate to decipher the

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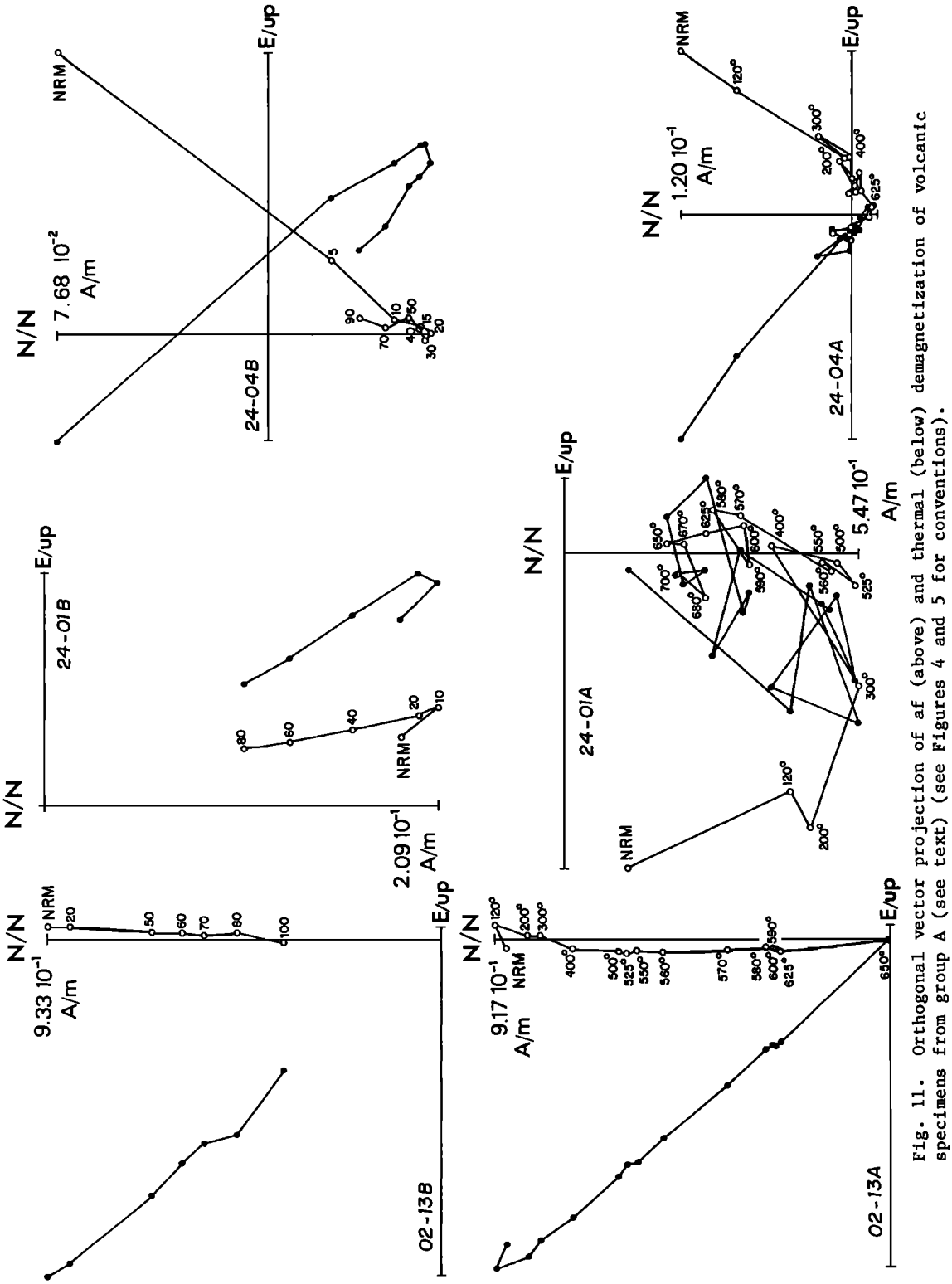


Fig. 11. Orthogonal vector projection of af (above) and thermal (below) demagnetization of volcanic specimens from group A (see text) (see Figures 4 and 5 for conventions).

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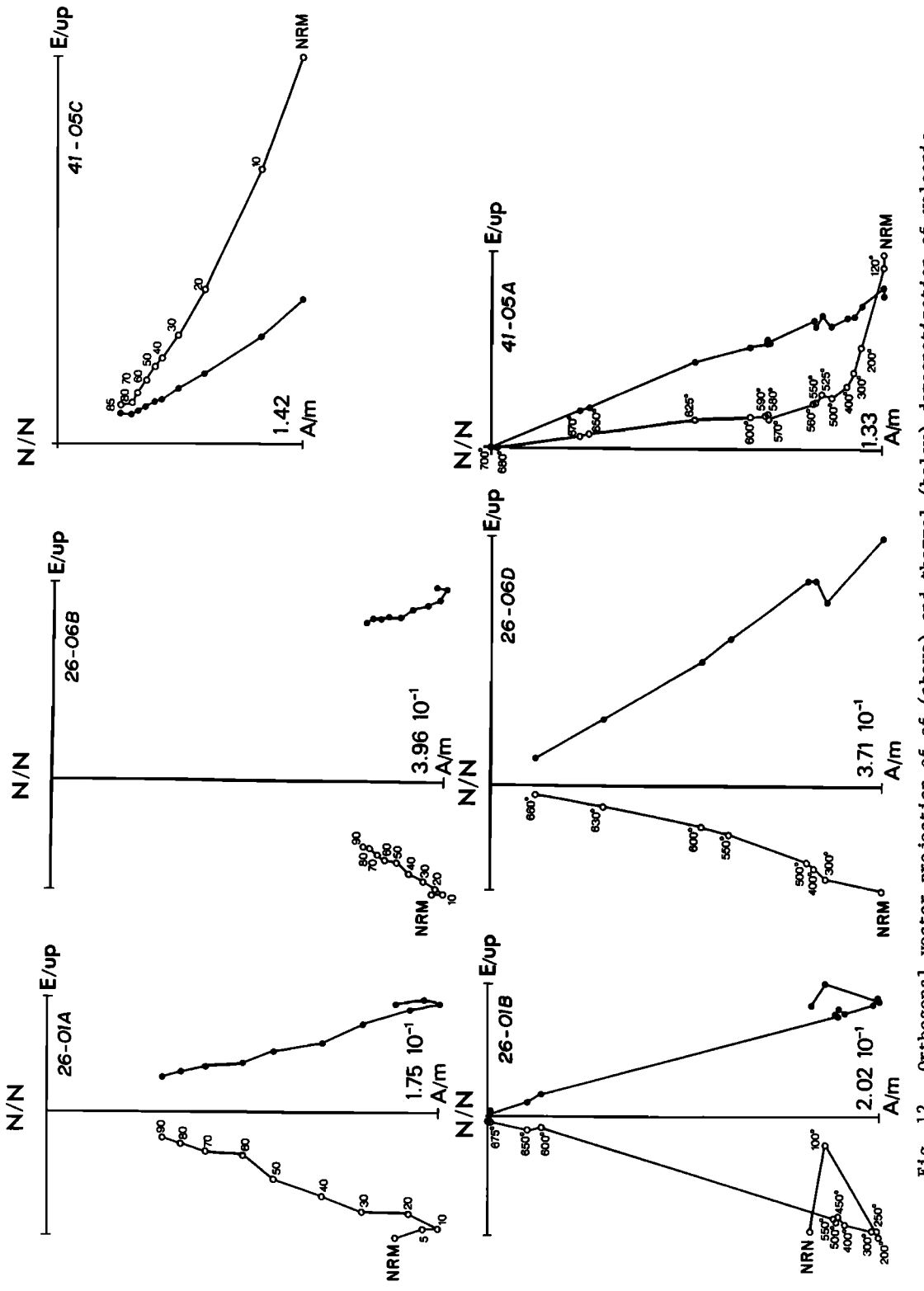


Fig. 12. Orthogonal vector projection of af (above) and thermal (below) demagnetization of volcanic specimens from group B (see text) (see Figures 4 and 5 for conventions).

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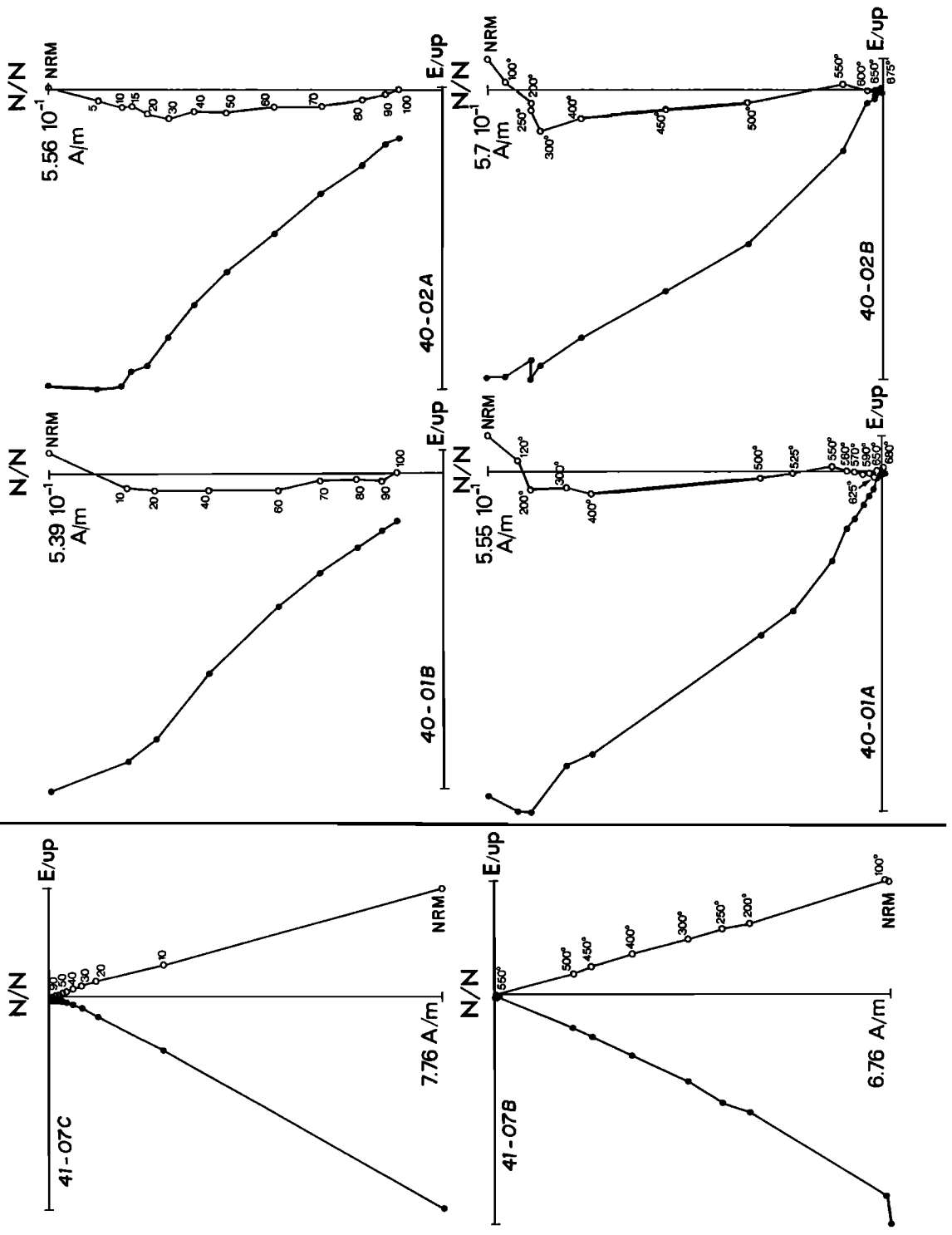


Fig. 13. Same as Figure 12. Site 40 reveals the superposition of a component with low blocking temperatures and high coercivities and a component with high blocking temperatures and low coercivities.

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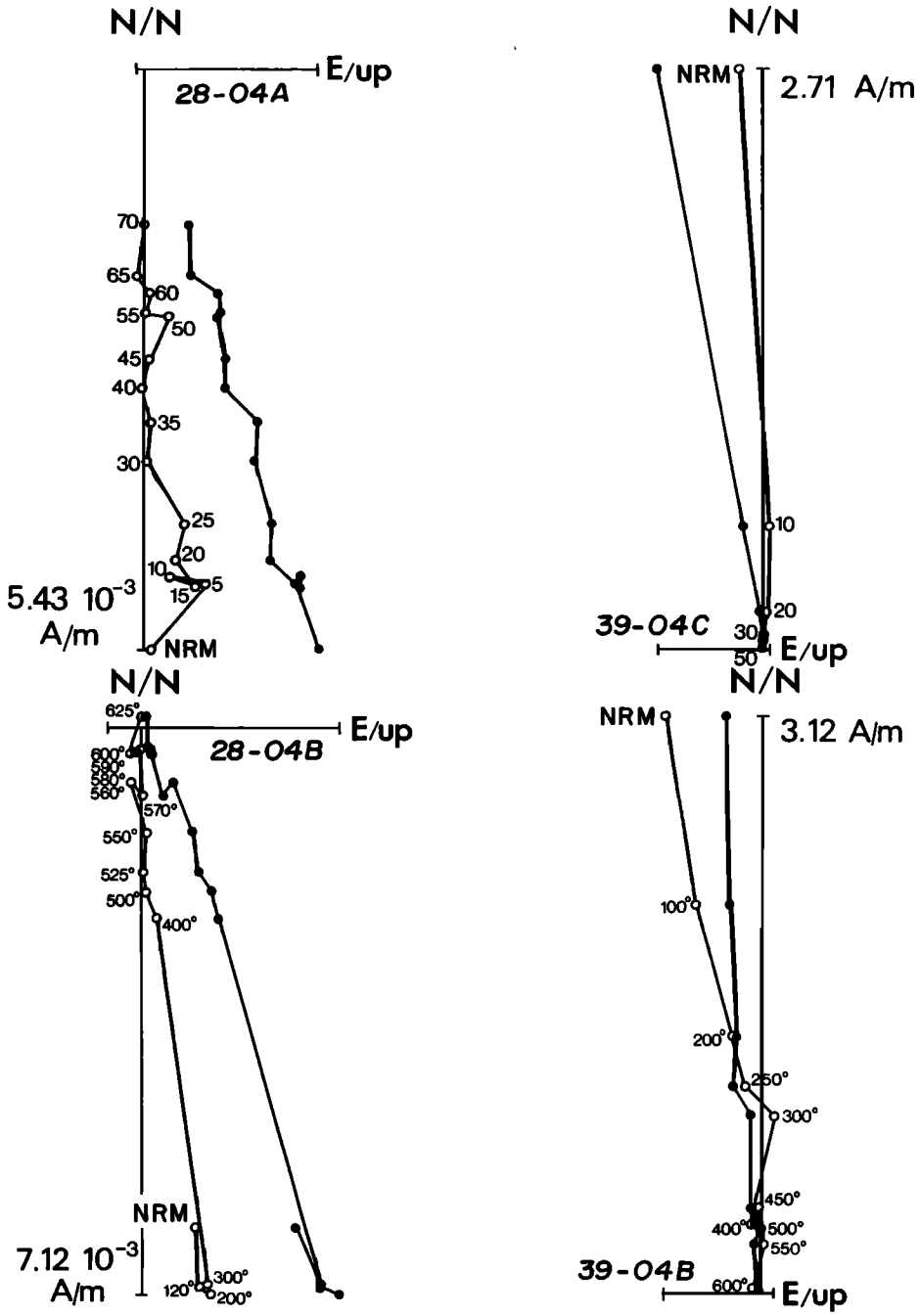


Fig. 14. Orthogonal vector projection of af (above) and thermal (below) demagnetization of volcanic specimens from group C (see text) (see Figures 4 and 5 for conventions).

magnetic record in sites 25 and 46 where rather noisy demagnetization curves are observed. To try and solve this problem, two specimens from each core have been separately analyzed in detail, one by af demagnetization up to 80 mT and the other by thermal treatment up to the Curie temperature of magnetite.

Since a second specimen was available from many cores in sites of groups A and B, both af and thermal treatments were applied to these samples in order to check for consistency. When consistent directions were obtained from several (two and sometimes three) specimens of the same core, a mean direction was computed for this core. This mean sample direction was then used to compute the site mean (the number of samples and the total number of specimens used in the mean are given in Table 4).

Results. Whenever more than one stable component of magnetization is identified in a site, the softer component turns out to be parallel to the present field direction as, for instance, in site 39 (before tectonic corrections are applied) with the noticeable exception of site 40 (see pilot study and below). This component, indicative of a recent remagnetization, is of no interest in this study and will not be included in the discussion. The results

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TABLE 4. Lingzizong Volcanics

Site	Polarity	N(n)	Geographic Coordinates D,deg	Geographic Coordinates I,deg	Stratigraphic Coordinates D,deg	Stratigraphic Coordinates I,deg	α_{95},deg	K
24	R+N	6(14)	151.7	-20.2	160.5	-30.9	13.6	25.1
25	N	4(5)	6.8	5.0	13.7	34.0	8.8	109.3
26	R	7(13)	207.1	-67.4	182.0	-28.3	5.9	105.8
28	R	7(14)	186.4	-23.0	180.0	-18.6	6.3	91.4
39	N+R	5(5)	341.4	44.6	346.4	15.7	21.5	13.6
40	N	6(6)	357.2	60.8	354.4	35.9	4.2	252.0
41	R	6(7)	167.8	-45.2	168.4	-20.9	5.4	152.8
47	R	5(5)	145.1	15.4	145.1	-14.5	9.6	64.7
Mean		8	170.1 (a)	-33.0	170.9	-25.5	11.0	26.4
Mean wo 47		7	174.8 (b)	-39.5	174.8	-26.7	9.5	41.3
46(granite)	R+N	3	178.5	-12.7	178.5	-12.7	12.2	103.3
Mean + 46		9	-	-	171.8	-24.1	10.0	27.3
2	N	10(15)	-32.9	46.1	-3.9	39.8	6.5	55.5

See Table 1 for symbols.
(a) α_{95}=23.6°, K=6.5 .
(b) α_{95}=20.5°, K=9.6 .

for each site are summarized in Table 4.

In site 24, at least three different flows have been sampled over a few hundred meters. The lower (older) flow, directly overlying the basal conglomerate of the Lingzizong formation, shows a normal polarity of magnetization. Despite the relatively small number of samples, the large α_{95} of this site (α_{95} = 13.5°) and the fact that three different flows with distinct polarities have been sampled suggest that most of the secular variation has been averaged out.

Site 27 leads to a mean direction very close to the present field in in situ coordinates. This site has been sampled in a dyke intruding the Takena, and thus no tectonic correction can be used to reject the suspicion of remagnetization. For this reason, site 27 has not been included in the computation of the mean Lingzizong direction.

In site 39, where several flows have been drilled, only five samples give a stable consistent direction of magnetization beside the present field component. However, two out of five samples show a reversed polarity. This suggests that the few intermediate directions observed in the site may indicate the occurrence of a reversal.

In site 40, two stable directions have been identified, none of them being close to the present field, either in geographic or in stratigraphic coordinates. Direction I corresponds to low blocking temperatures and high coercivities and direction II to higher blocking temperatures and lower coercivities (see pilot study and Figure 13). Before correcting for bedding, direction II (D = 76.5°, I = 84°, α_{95} = 9°, K = 109) gives surprisingly steep inclinations which rules out any posttilting remagnetization origin. This direction still shows high inclinations in stratigraphic coordinates (D = 3.5°, I = 64°; α_{95} = 9°, K = 109), but it now comes closer to the present field, although remaining significantly different from it. It is remarkably well defined in all samples both by af and thermal treatment. Nevertheless, no straightforward interpretation can be given for this direction. Direction I is significantly different from the present field direction before tectonic corrections are applied and gives shallower than present inclinations when these corrections are applied. Thus it will be considered as a primary direction and included in the mean (Table 4).

In site 41, three cores from the top flow (41-07, see pilot study, 41-08, 41-09) lead to anomalous directions. In addition, these samples show, unlike the rest of the site, a range of blocking temperatures characteristic of the presence of magnetite. The directions of these samples, although very consistent within a single core, are quite scattered about the mean direction of the flow (α_{95} = 36°). This mean (D = 137°, I = 27°) does not compare with the mean of all other flows of the site (see Table 4). These directions may be interpreted as being due to the occurrence during cooling of the flow of a "far-sided R to N" reversal [Hoffman and Fuller, 1978]. However, as we have noted in the pilot study, this flow stands at the top of a hill, and the anomalous component could also be due to lightning. These three core directions are not included in the computation of the mean direction of the site.

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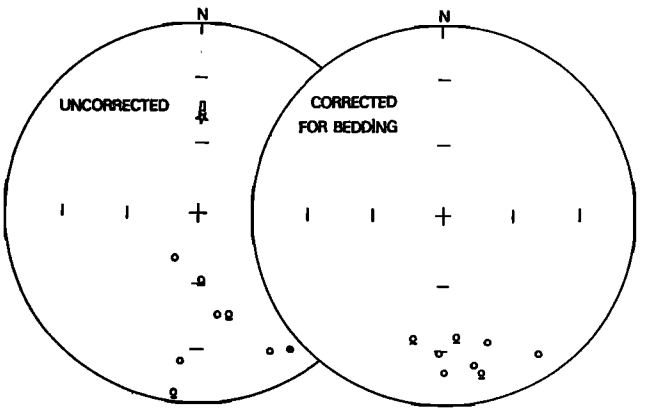


Fig. 15. Equal-area projection of mean site directions for the Lingzizong volcanics before and after correcting for bedding. Normal directions have been reversed (underlined). Open square, present field direction; star, present dipole field direction.

The global mean of eight sites from the Lingzizong formation (24, 25, 26, 28, 39, 40, 41, and 47) is computed in geographic and stratigraphic coordinates (Table 4). Four sites show a reversed polarity. Two sites with normal polarity and two with mixed polarities have been reduced to common reversed directions in Figure 15. This volcanic formation leads to a fold test which is significant at the 99% probability level (K increases from 6.5 to 26.4; the test is significant at this probability level for a K ratio of 3.7) and therefore indicates a prefolding magnetization of the Lingzizong with direction D = 171°, I = -25.5° (or, equivalently, D = 351°, I = 25.5°; α_{95} = 1°, K = 26).

Site 46 has been sampled in the Lhasa granite. Tectonic corrections are not known for this body, and the interpretation of its paleomagnetic direction can only be limited. Only three samples give a stable direction of magnetization, others being probably too altered. The observation of a reversal (one normal direction is antiparallel to the two others) gives the mean direction for this site more reliability (although it could be a self-reversal phenomenon). This granite has been dated at 53 Ma [Schärer et al., 1984] by a U-Pb method on zircons and was therefore intruded during the emplacement of the Lingzizong. Since tectonic corrections are likely not to exceed 20°-30° (mean value of the dips of the volcanic flows in the neighborhood of the granite), we tentatively calculate an overall mean including the eight sites of Lingzizong and this granite (Table 4). The mean direction obtained D = 352°, I = 24° (α_{95} = 10°, K = 27) is not significantly different from the previous mean Lingzizong direction.

Site 2, near Nagqu and north of the Nyainqentanglha, does not belong to the Lingzizong formation. This site consists of four different series of andesite and tuff flows. The total thickness sampled is of the order of half a kilometer. Samples from the lower series of flows show only the present field direction, in geographic coordinates. The same present field component is observed in the upper series of flows (see Figure 11, 02-13B and 02-13A between 120° and 400°), but it overprints a more stable component. This latter component gives a mean direction D = 356°, I = 40° (α_{95} = 6.5°, K = 55) which compares with the mean Lingzizong direction in declination but shows a singificantly steeper inclination.

6. Analysis of the results

In Table 5, we have summarized all the paleomagnetic directions derived in this paper (see also Figure 16). In the Takena formation we have observed a single component of magnetization in the north of the Lhasa block (or Southern Tibet) (referred to below as direction N) and two components in the south (HT and LT). The Lingzizong volcanics revealed a single direction of magnetization. In addition, a preliminary direction has been derived for the older volcanics of site 2, possibly associated with the Bangong-Nujiang suture zone. The consequences and relationship between these different directions are analyzed in this section.

Primary Magnetization of the Takena Formation

Magnetization components N (Figure 8), north of the Nyainqentanglha, and HT (Figure 9), in the southern part of the block, successfully pass a fold test at the 99% probability level. Therefore these components clearly predate the folding of the Takena formation. The age of folding is not constrained accurately but must predate the oldest age available for the unconformably overlying Lingzizong formation, i.e., 60 Ma [Maluski et al., 1982].

The occurrence in red beds of nearly antiparallel components of magnetization has been discussed by Larson et al. [1982]. These authors attribute this occurrence to a long-term acquisition of chemical remanent magnetization (CRM): under oxidizing conditions that are likely to have prevailed at the time of deposition of the Takena the original depositional remanent magnetization (DRM) may be replaced, through a process of authigenesis, by a CRM acquired over several millions of years. Larson et al. [1982] propose that this process is to a large extent controlled by lithology. We observed that site 12, sampled in coarse-grained sandstone, shows results which are distinct from all other sites from groups I and II sampled in finer-grained sandstones. Sample 12-05A (Figure 4a) revealed an intermediate direction above 680°C which may very well result from the superposition of two nearly antiparallel components with high blocking temperatures and similar intensities.

The fact that the more stable remanence of the red beds shows a systematic normal polarity is consistent with such a mechanism having occurred during the Cretaceous long normal polarity interval (110-80 Ma), which indeed coincides with deposition of the Takena. Furthermore, the occurrence of a minor superimposed antiparallel component observed in many sites would imply that acquisition of magnetization may have continued until the end of the long normal interval, i.e., 80 Ma. The fact that this component is less observed in the southern part of the Lhasa block may imply that the acquisition of magnetization may have been faster than in the north.

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TABLE 5. Summary of Magnetic Directions in Southern Tibet

Formation-Age	N	D,deg	δD,deg	I,deg	p,°N	δp	α_{95},deg	K	K ratio	VGP λ,°N	VGP φ,°E	VGP A_{95},deg	VGP K
Takena Northern region (N direction) Albo-Aptian	6	-21.3	9.8	25.4	13.3	5.1	8.9	57.0	6.2	63.5	325.4	6.5	106
Southern region (HT direction) Albo-Aptian	8	-5.7	9.0	22.6	11.8	4.7	8.3	45.7	19.9	71.2	288.4	7.9	50
Mean N + HT (a)	14	-12.3	7.2	24.0	12.5	3.8	6.6	37.3	11.0	68.8	306.3	6.2	43
Mean N + HT (rotated)	14	-21.3	6.0	23.8	12.4	3.1	5.5	53.1	-	63.4	323.2	4.8	69
Mean inclination				23.6	12.3		(σ=7.2)						
Lingzizong (60-45 Ma)	8	-9.1	12.2	25.5	13.4	6.4	11.0	26.4	4.1				
Takena Southern region (LT direction) (60-45 Ma ?)	8	-2.9	21.6	21.8	11.3	11.2	20.1	8.5	2.2				

p : paleolatitude ; δp : uncertainty on the paleolatitude = $\alpha_{95}(1+3\cos^2 p_0)/2$ where p_0 is the paleocolatitude; δD : uncertainty on the declination = $\alpha_{95}/\cos I$; see Table 1 for other symbols.

(a) the declination difference ΔD between components N and HT is 15.6°±13.3°; the differential rotation ΔR ,computed from VGPs at the point (91° E, 31° N), is 16.2°±13.3° . The uncertainties on ΔD and ΔR are computed after $(\delta X_1^2 + \delta X_2^2)$.

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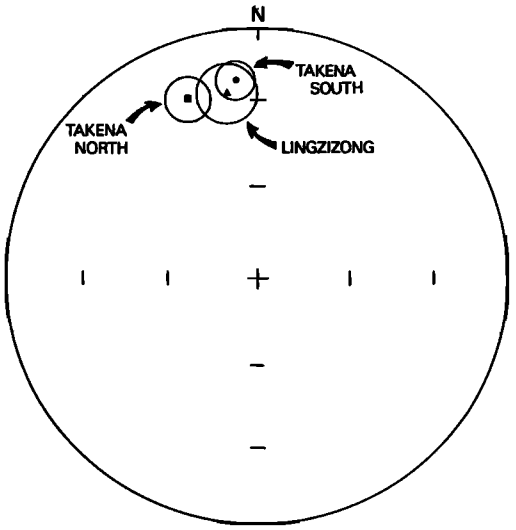


Fig. 16. Equal-area projection of the mean primary directions of magnetization of the red beds of the Takena formation, north of the Nyainqentanglha range (direction N) and south of this range (direction HT), and of the mean primary direction of magnetization of the Lingzizong volcanics.

These observations are all consistent with mean components N in the north and HT in the south being coeval CRM acquired sometime between 110-100 Ma (deposition) and 80 Ma (end of the Cretaceous long normal interval). Indeed, the similar paleolatitudes for the two parts of the Lhasa block (Table 5), 13.5° ± 5° N in the north (N component) and 12° ± 4.5° N in the south (HT component) are consistent with these two magnetic components having been acquired at the same time.

On the other hand, a significant difference in declination is observed between the two directions N and HT (Table 5 and Figure 16):the northern part has undergone a 21.5° ± 10° counterclockwise rotation with respect to true north, whereas the southern part has only undergone 5.5° ± 5° of the same rotation. This declination difference implies 16° ± 13° of relative rotation between the two subunits since the acquisition of their magnetization.

Finally, we wish to determine as acccurately as possible the mean latitude of the entire Lhasa block at the time of magnetization of the Takena formation. This can be done in three different ways. We can compute an overall mean of all site directions N and HT (Table 5). The resulting direction D = 347.5°, I = 24° has good statistical parameters (α_{95} = 6.6°, K = 37) and retains the positive fold test at a probability level much better than 99% (K ratio of 11 for 14 sites (see Table 5) the fold test being significant at the 99% level when the K ratio is 2.55). However, the previous discussion of declination differences on both sides of the Nyainqentanglha suggests that we should rotate one block with respect to the other: if the directions of the southern sites are rotated counterclockwise by the exact amount of the mean declination diffe-rence (Table 5: mean N + HT rotated), the incli-nation remains essentially the same (I = 23.8° compared to 24.0°) and the 95% confidence limit is reduced to 5.5°, as expected. A third estimate can be obtained with inclinations only, assuming a Gaussian distribution, and again the same value I = 23.6° results from the analysis (Table 5).

The final estimate of the early Late Cretaceous paleolatitude of Southern Tibet (for a reference location now near Lhasa) is 12.5 ± 5° N. The uncertainty is remarkably small despite a possible long-term acquisition of the CRM.

Primary Magnetization of the Lingzizong Formation and Remagnetization of the Takena Formation

The final mean direction of magnetization of the Lingzizong formation (Table 5) has been shown to predate the folding of the volcanics. However, a certain degree of scatter remains after tectonic corrections are applied (α_{95} = 11°, K_S = 26, K_S/K_G = 4.1; see Table 5 and Figure 15), in particular, in declination. This dispersion may be attributed to several factors. First, it may be due to slightly improper tectonic corrections resulting from inaccurate bedding measurement in the field. It is, indeed, sometimes difficult to recover the original level of deposition in some volcanic bodies (such may particularly be the case for the ignimbrite of site 40 and the andesite of site 47). Second, it may be due to the secular variation which may not have been averaged out in all sites, as, for instance, sites 25 and 47 for which only a few (less than five) independant directions have been obtained (indeed, these site mean directions fall quite far from the mean Lingzizong direction, see Figure 15). However, these two phenomena cannot result in the anisotropic scatter which is observed. As a third possibility, apparent polar wander may have been recorded in this formation. The Lingzizong is likely to have been deposited over a large period of time spanning 10-20 Ma, as is indicated by the two absolute ages available (59 ± 2 Ma and 48.5 ± 1.5 Ma) and the Lhasa block may have undergone latitudinal displacements and/or local rotations. However, very little dispersion is observed in the inclinations, and most of the spread is due to declination variations. It thus suggests that local rotations may have occurred during or after the emplacement of the Lingzizong volcanics.

The LT component of magnetization of the red beds observed in the southern part of the Lhasa block has been shown to postdate in part the folding of the Takena formation (the fold test is not positive at the 95% confidence level). This suggests that LT may be a remagnetization component: it is not distinguishable at the 95% probability level from the direction of magnetization of the Lingzizong (see Table 5) and is only observed in areas where the Takena is overlain by the younger Lingzizong. In addition, both normal and reversed polarities are observed. Thus LT is likely to be associated with the emplacement of the volcanics.

In Figure 10 we have plotted the effect of the tectonic corrections on the LT directions for each site. The root mean square value of the dip of the volcanic flows in the south of the Lhasa block is of the order of 25°, whereas it is above 50° for the Takena red beds. For the latter formation, however, the intensity of folding tends to be much higher between Lhasa and

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Yanbajain (near 7°) than near Linzhu (only 3°). It appears, then, that approximately one third of the total folding of the Takena formation is accounted for by the younger tectonic phase which also involved the Lingzizong volcanics. Indeed, we observe on Figure 10 that overcorrections vary from about 30% (for the sites near Linzhu) to 60% (for the sites near Lhasa-Yanbajain) of the total tectonic corrections of each site. This further substantiates the hypothesis that the LT direction recorded in the Takena has been produced by the emplacement on top of it of the Lingzizong volcanics.

We observe that about 45% of the Lingzizong volcanic sites have a normal polarity (this value being consistent with the ratio of normal to reversed periods during Paleocene-Eocene time [see Lowrie and Kent, 1983]), whereas 75% of the LT directions for Takena sites have this polarity. An alternate interpretation would involve a first remagnetization event in a normal field (affecting sites 23, 32, 33, 34, 35, and 38), likely to have occurred before the end of the Cretaceous long normal interval and thus prior to the extrusion of the Lingzizong, followed by a second remagnetization event in a reversed field, associated with the Lingzizong but only observed in the most tectonized outcrops of the Takena (sites 29 and 30). However, this interpretation requires complicated assumptions on the tectonic history of the Takena and would not explain the absence of LT in the north of the Lhasa block. We suggest that the discrepancy on the polarity ratios is better explained by a bias in the sampling of the two formations. Indeed, the number of independant sites is too small for the polarity ratios to be reliable. Furthermore, it can be observed in Tables 3 and 5 that the mean LT direction computed from the six normal sites only cannot be distinguished from the mean Lingzizong direction and even falls closer to it than the overall mean LT direction which includes the two reversed sites.

In any case, the direction of magnetization of the Lingzizong implies that the Lhasa block was situated at a paleolatitude of 13.5° ± 6.5° N at the time of extrusion of the volcanics, i.e., between 60 and 50 Ma. It further shows that no significant net rotation of the block has occurred since that time.

Comparison With Previous Paleomagnetic Studies

The direction of primary magnetization of the Takena formation derived in this paper differs significantly from previous determinations. The discrepancies with the preliminary results of Chinese investigators (D = 338°, I = 40° [Zhu et al., 1981] and D = 335°, I = 16° [Zhu et al., 1977]) can be explained by insufficient demagnetization of the red beds. Indeed, in these two papers the samples are reported to have been submitted to alternating field demagnetization only. We have shown here that such a treatment was often insufficient to reveal the primary magnetization in the Takena. This was already clear from Zhu et al. [1981] in which Figure 3 shows that less than 10% of the original remanence is destroyed by such a treatment.

A more recent determination of Takena paleomagnetic directions was obtained by Westphal et al. [1983] (see also Pozzi et al. [1982]) during the first joint French-Chinese expedition to Tibet in 1980. This determination (D = 33°, I = 38°; α_{95} = 8°) would imply a much higher paleolatitude (21° ± 8° N) than that proposed in this paper. The discrepancy is due to incomplete thermal demagnetization of some samples. This incomplete isolation is well understood if one recalls that sampling during the 1980 campaign was restricted to the southernmost part of the Lhasa block where folding and remagnetization by the overlying Lingzizong volcanics are known to be most severe.Indeed, the treatment has been applied to the rocks up to a temperature of 650°-665°. We have shown above in this paper that the primary component of the Takena in the southern part of the Lhasa block (component HT) was, in some instances, still not isolated at 650°C and that demagnetization had to be performed all the way up to the Néel temperature of hematite. Thus we suggest that some of the directions used by Westphal et al. [1983] to derive the mean primary magnetization of the Takena may not always be truly of primary origin but rather result from mixing with a steeper secondary component (component LT). This is particularly clear in Figure 3a of their paper on sample 5-8c from site 5, and indeed, this site gives the steepest value for the inclination and is likely to affect the computation of the mean direction. Ambiguities about a potential partial remagnetization hypothesis raised from the comparison of the VGP of the sites with the APWP of India [Pozzi et al., 1982] may originate from incomplete isolation of the true primary component.

On the other hand, the mean paleomagnetic direction obtained in the present paper for the Lingzizong volcanics is in good agreement (declination, inclination, polarity) with the two site directions (sites 3 and 7) proposed by Westphal et al. [1983] for this formation.

7. Interpretation

The previous section summarizes the constraints which can be placed on the Late Cretaceous and younger tectonic history of the Lhasa block by paleomagnetic directions recorded in the Takena and Lingzizong formations. These constraints concern (1) the latitudinal positions of the block at key periods, (2) the extent of tectonic phases within the block, and (3) the discovery of previously undetected differential rotations within the block. In this section, we analyze the consequences of these observations, compare them with available geoscience data from Southern Tibet, and propose a synthesis of the evolution of this microcontinent in the last 100 Ma.

Paleolatitude of Southern Tibet and Crustal Shortening of Eurasia

A fundamental consequence of the uniformity of inclinations of the primary magnetization of Takena red beds over the whole extent of the Lhasa block is that the block had already formed a single tectonic unit by middle Cretaceous (Albian-Aptian) time. Since no significant

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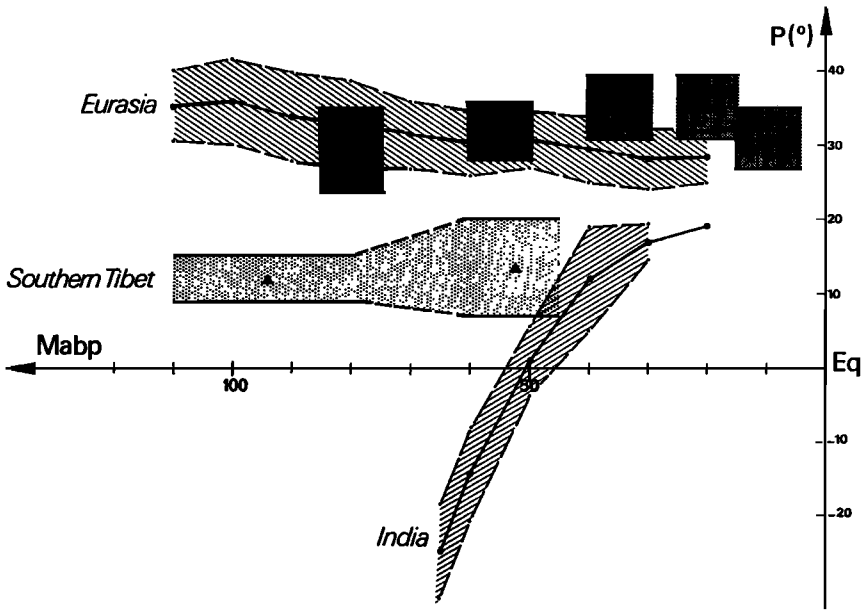


Fig. 17. Paleolatitude of Southern Tibet since middle Cretaceous and before the India-Asia collision, as deduced from paleomagnetic data (middle curve). It is compared with the paleolatitude expected at a point near Lhasa (91° E, 30° N) considered to have drifted with India (lower curve) or Eurasia (upper curves). The curve for India is computed after the APWP for this continent [Klootwijk and Radhakrishnamurty, 1981]. The continuous curve for Eurasia is deduced from the APWP of this continent of Irving [1977]. The second curve for Eurasia is computed from a series of independant Upper Cretaceous and Cenozoic poles [Achache et al., 1983a]. The collision between India and Southern Tibet is shown to occur near 50 Ma but appears to rely strongly on the age of the Cenozoic Lingzizong volcanics. The paleolatitude uncertainties are computed by $\delta p = \alpha_{95}(1+3\cos^2 p_o)/2$, where p_o is the paleocolatitude, for Southern Tibet, and by $\delta p = A_{95}$ for Eurasia and India.

crustal shortening has occurred within the block since then, the existence of a younger suture zone within the block is ruled out. This is consistent with recent field observations that the ophiolite outcrops near lake Nam Co are in fact klippen of nappes, possibly originating from the northern Bangong-Nujiang suture zone [Marcoux et al., 1982]. Considering the paleomagnetic uncertainties, it is almost a surprising coincidence that the 1.5° difference in mean paleolatitudes on both sides of the Nyainqentanglha is equal to the present-day latitudinal mean distance between the northern and southern parts of Southern Tibet (see Figure 2).

Previous geological data [Chang and Cheng, 1973], supplemented by recent geochemical and radiometric studies [Allègre et al., 1984], imply that suturing between Northern and Southern Tibet along the Bangong-Nujiang suture zone was completed before deposition of the red beds, i.e., prior to 110 Ma. All other sutures farther north are considered to be even older [Chang and Cheng, 1973]. Therefore the southern edge of the Lhasa block was at the same time the southern margin of continental Eurasia by middle Cretaceous. This is consistent with the paleontological observations that the Takena red beds were deposited in a deltaic environment [Jaeger et al., 1982]. The mean paleolatitude of Southern Tibet, as determined from the Takena primary magnetization, is 12.5° ± 3° N. This places the southern margin of Eurasia at 11.5° ± 3° N (for a reference location on the Yarlung-Zangbo suture zone). The younger Lingzizong pole puts this margin at 12.5° ± 6.5° N and thus indicates that Southern Tibet remained at a constant paleolatitude throughout the Upper Cretaceous and the Paleocene.

This is compared in Figure 17 to the paleolatitude which would be expected if Southern Tibet (actually, a reference location at 91° E, 30° N) had remained in a stable position with respect to Eurasia. Two different determinations of the APWP of Eurasia are used for this comparison [Irving, 1977; Achache et al., 1983a]. Irving's curve is essentially valid for northern Eurasia, whereas the Upper Cretaceous pole of Achache et al. [1983a] has been derived with special emphasis on eastern and central parts of Eurasia. The two sets of data are, however, consistent for the geological periods of interest in this paper, i.e., between 90 and 40 Ma. We note a large discrepancy between the expected and observed paleolatitudes at 80-90 Ma: 21° ± 6.5° with Irving's [1977] pole and 17° ± 7.5° using the Achache et al.[1983a] determination. This discrepancy remains essentially the same around 50 Ma, when the Lingzizong volcanics are erupted : 17° ± 7.5° with Irving's curve and 18.5° ± 7.5° with Achache et al.'s Paleocene/Eocene pole. The absence of latitudinal motion of the Lhasa block between 100 and 50 Ma is therefore mirrored in the absence of such motion for stable Eurasia: no north-south shortening occurred within Eurasia throughout this period. The discrepancy in paleolatitudes disappears

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during the last 50 Ma and thus implies large-scale continental crustal shortening north of the Lhasa block. This north-south shortening amounts to 1900 ± 850 km when computed with the Takena pole and 2050 ± 850 km with the Lingzizong pole, using Achache et al.'s [1983a] Eurasian poles. This figure is on the high side of previous estimates made by Molnar and Tapponnier [1975] and Tapponnier et al. [1982,1984]. Although details of the shortening process are beyond the scope of this paper, the change in position of the southern margin of Asia after a long period of stability is seen to coincide with the onset of the India/Asia collision. This large shortening may be distributed over a long distance, as far north as the Tien Shan, and may be accounted for by a number of major strike-slip faults [Molnar and Tapponnier, 1975; Tapponnier et al., 1982,1984] and/or by some crustal thickening by folding and intracontinental thrusting [Allègre et al.,1984; Hirn et al., 1984].

It may be worthwhile to point out that preliminary paleomagnetic results from Jurassic rocks in the Lhasa block already indicate rather low latitudes [Zhu et al., 1981] (this needs confirmation in view of the discussion of demagnetizations made in this paper). Moreover, paleontological observations [Allègre et al., 1984; Jaeger et al., 1982] suggest that the Lhasa block was already in northern tropical latitudes in the late Permian. Therefore the Southern Tibet microcontinent may have remained at roughly constant equatorial to northern tropical paleolatitudes over the entire Mesozoic era. If the block was ever part of Gondwana, it had separated from it at least by early Permian [Allègre et al., 1984].

In the Upper Cretaceous, India was still far in the southern hemisphere (see bottom curve of Figure 17, representing the paleolatitude of the same point near Lhasa, at 9°E, 30°N, deduced from the Indian APWP of Klootwijk and Radhakrishnamurty [1981]. Even at 50 Ma, when the collision had probably already started in Tibet [Patriat and Achache, 1984] a latitude difference of 12.5° ± 1° remains between Southern Tibet and the northern margin of India (taken along the Zangbo suture). This estimate relies heavily on accurate age determinations of the Lingzizong volcanics, which are at present still too scarce, and on an accurate evaluation of the Indian APWP, which relies for most of the Cenozoic on Deep Sea Drilling Project (DSDP) cores which are prone to errors and require careful evaluation [Klootwijk and Radhakrishnamurty, 1981]. The latitude discrepancy implies several hundred kilometers of convergence and crustal shortening between the Lhasa block and the stable Indian craton since the Eocene. This shortening may result from subduction of a limited amount of continental crust along the Zangbo suture zone and from underthrusting of continental crust along major thrust faults such as the Main Boundary Thrust, Main Central Thrust, and Kangmar thrust, south of the Zangbo [Allègre et al., 1984; Burg, 1983; Brunel, 1983; Hirn et al., 1984]. Ongoing paleomagnetic research in Tibet, in particular, the study of Paleocene samples from Dingri (between the high Himalayas and the Zangbo [Besse et al., 1984]), and the careful analysis of magnetic anomalies in the Indian Ocean [Patriat and Achache, 1984] provide independant and better constrained estimates of this shortening.

Finally, it should be emphasized that no significant latitudinal displacement of Southern Tibet is indicated by the data derived in this paper, prior to the collision (Figure 17). Thus the model of southward movement of Tibet during the Paleocene, proposed by Westphal and Pozzi [1983], appears not to be sustained by the present data.

Compressive Tectonic Phases in Southern Tibet

The age of the major phase of folding of the Takena formation is still uncertain, and its origin poses a vexing problem. This phase is rather intense and well recorded in the southern part of the Lhasa block, and the unconformably overlying Lingzizong formation constrains its age to be older than 60 Ma [Tapponnier et al., 1981a; Maluski et al., 1982]. The positive fold test on the primary magnetization of the Takena implies that folding is probably younger than 80 Ma, since acquisition of pretectonic CRM may have continued until the end of the Cretaceous long normal interval. Folding of the Takena north of the Nyainqentanglha is less severe than in the south but is most likely due to the same phase.

The age of this tectonic phase, between 80 and 60 Ma, cannot be associated with the collision of India, which could not have started until 60 Ma ago. As an alternate tentative solution, Allègre et al.[1984] have suggested that an island arc may have collided with Southern Tibet, but no evidence of remnants of this arc have been found in the Gangdise until now [Coulon et al., 1982]. Another mechanism is provided by the collision of an oceanic plateau or of a large aseismic ridge or seamount province. Using paleoreconstructions of the Indian Ocean [Patriat, 1983], Patriat and Achache [1984] associate this phase, the major phase of folding to occur over the entire extent of the Lhasa block, with the collision and subduction just prior to anomaly 29, that is, 65 Ma ago, of an aseismic volcanic ridge (or plateau) generated by the Réunion hot spot. Indeed, folding of the Takena formation has been described as corresponding to "Andean" subduction [Burg, 1983], and it has been suggested that the "Andean" orogenies may be related to such collisions of microblocks or oceanic plateaus [Nur and Ben Avraham, 1981; P. Tapponnier, personal communication, 1984].

A second phase of folding of the Takena formation occurred in the south of the Lhasa block and is also recorded by the overlying Lingzizong formation [Burg, 1983]. This phase postdates both the emplacement and the magnetization of the Lingzizong volcanics and postdates or is partly coeval with the remagnetization of the underlying red beds. Its age must then be younger than about 45 Ma (48 Ma age determination by Westphal et al. [1983]). The intensity of folding appears to be more severe in the Lhasa-Yangbajain area, where bedding planes are sometimes vertical and even slightly overturned, than in the Pengpo-Linzhu area (Figure 2). Therefore the intensity of folding appears to increase from east to west in marked departure with the east-west cylindrical symmetry

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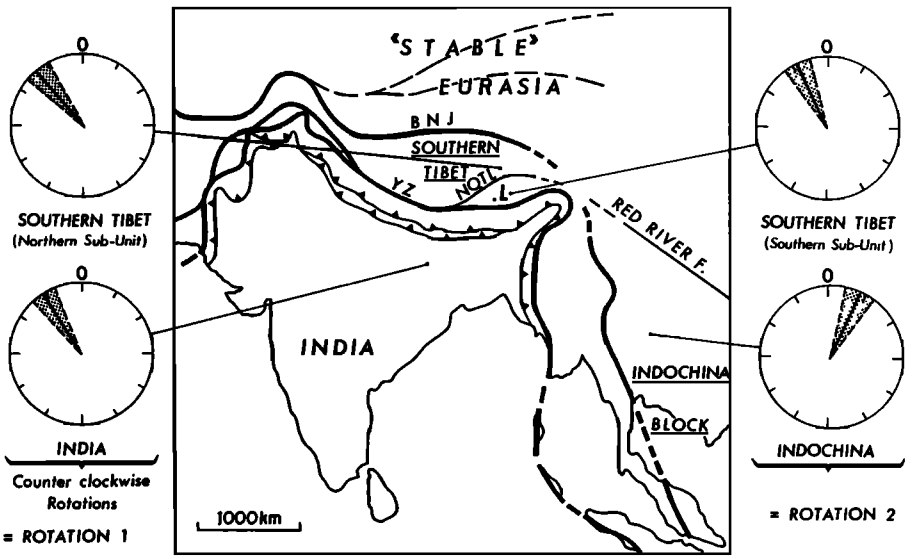


Fig. 18. Sketch map of south-central Eurasia indicating the major tectonic units surrounding Southern Tibet. Heavy lines, suture zones (BNJ, Bangong-Nujiang; YZ, Yarlung-Zangbo); barbed lines, continental thrusts. Also indicated are the Nyainqentanglha (NQTL) and the Red River fault zones. Circles on the left show the counterclockwise rotations of India and Southern Tibet with respect to "stable" Eurasia. The two upper circles indicate the differential rotation within Southern Tibet compatible with left-lateral displacement along the NQTL. Also shown is the clockwise rotation of the Indochina block with respect to stable Eurasia associated with left-lateral displacement along the Red River fault [Tapponnier et al., 1982]. Although suggested by the rotations, the continuity of structures (faults and sutures) has been voluntarily omitted near the Assam syntaxis. L, Lhasa. The rotation uncertainties are computed from $\delta D = \alpha_{95}/\cos I$ for Southern Tibet, and from $\delta D = \sin^{-1}(\sin A_{95}/\sin p_o)$ where p_o is the paleocolatitude, for Eurasia, India and Indochina.

which is often assumed to occur throughout Tibet. This and the fact that this second phase is clearly not observed in the north of the Lhasa block (see above, section 2) points to a potential link with the activation of the Nyainqentanglha fault system. In any case, this phase is clearly related to the ongoing collision of India with the Lhasa block and the rest of Asia.

Figures 9 and 15 (and to a lesser extent Figure 10) show that the mean directions of magnetization of individual sites recorded independently in the Takena red beds and the Lingzizong volcanics, south of the Nyainqentanglha, present a consistent spread in declination. This could be indicative of the occurrence in the southern part of the Lhasa block of a certain amount of penetrative deformation [J.P. Brun, personal communication, 1984]. This observation deserves further studies, and part of the paleomagnetic sampling during the 1982 campaign in Tibet was devoted to this problem [work in preparation].

Cenozoic Tectonic Rotations in Southern Tibet

The direction of primary magnetization of the Takena red beds from the northern part of the Lhasa block implies a significant counterclockwise rotation of the block, younger than 80 Ma as discussed above (Table 5 and Figure 16). In the same time, since Upper Cretaceous, Eurasia has continuously rotated clockwise. The expected "Eurasian" declination at our reference point near Lhasa (91°E, 30°N) is 20° ± 8° (computed from the Upper Cretaceous pole of Achache et al. [1983a]), whereas the corresponding declination in the northern part of Southern Tibet is -21.5° ± 10° (Table 5). The resulting counterclockwise rotation of this unit with respect to Eurasia has the quite large value of 41.5° ± 12.5° (Figure 18).

The APWP of India records a similar counterclockwise rotation during the Cenozoic. Although the APWP entirely relies on DSDP cores in the last 60 Ma [Klootwijk and Radhakrishnamurty, 1981], a Deccan Trapps mean paleomagnetic pole dated at 60-70 Ma [Klootwijk, 1979] implies a counterclockwise rotation of India with respect to stable Eurasia which amounts to 35° ± 10.5° (Figure 18). This rotation is also apparent in the record of marine magnetic anomalies from the Indian Ocean [Molnar and Tapponnier, 1975; Patriat et al., 1982; Patriat, 1983]. These anomalies demonstrate a rather sudden change in the direction of drift of India near the time of anomaly 20 (44 Ma), following a major decrease in velocity starting at the time of anomaly 22 (50 Ma) [Patriat and Achache, 1984]. Joint use of Indian and Atlantic ocean anomalies constrains the relative motion of India with respect to Eurasia and shows the counterclockwise motion of India to be a rather continuous phenomenon with a faster early phase between anomalies 20 and 13 (i.e., between 44 and 36 Ma [Patriat and Achache, 1984]). Thus this rotation appears to be linked with the collision. The timing, amount, and sense of the rotation are also remarkably consistent with those in the northern part of the Lhasa block, favoring strongly the idea of a causal

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relationship between the two. In addition, at the time of the collision, Southern Tibet and Eurasia had already been sutured for tens of millions of years. Thus only a major tectonic event, such as the collision of India, can have induced such a large relative rotation between these two units.

This differential rotation implies large right-lateral motions north of the Lhasa block. It could for instance be accounted for by right-lateral strike slip along the Bangong-Nujiang suture. Strike-slip rejuvenation of a preexisting suture may be one of the features that characterize the evolution of the continental mosaic [e.g.,Tapponnier et al., 1982; Allègre et al., 1984]. This has been documented by field evidence on the Zangbo suture [Tapponnier et al., 1981a, 1982] and is supported both for the Zangbo and Bangong-Nujiang sutures by available seismic data [Hirn et al., 1984]. However, there is as yet no field evidence for the sense of motion on the Bangong-Nujiang zone near Anduo. The sense which results from this paleomagnetic analysis is the opposite of that observed by Peltzer et al. [1982] in their indentation experiments. We attribute this to the rotation of India, which is not yet modeled in these indentation experiments and which results in competing senses of motion for the extruded blocks: counterclockwise rotation in sympathy with India versus southeastward extrusion and clockwise rotation. Achache et al. [1983a] noted this and pointed the need for experiments with a rotating indenter. Further paleomagnetic data and structural field work in Northern Tibet will also be necessary to resolve this issue.

The difference in declination between the southern and northern parts of the Lhasa block across the Nyainqentanglha range amounts to 15.5° ± 13.5° (Table 5). The corresponding difference between the southern part of the Lhasa block and India across the Zangbo suture is only 11.5° ± 1°. These differences can be interpreted either in terms of rigid block rotations or in terms of penetrative deformation. The first hypothesis implies left-lateral shear on the Nyainqentanglha, which does not seem to agree with the sense proposed on the basis of field observations [Mattauer et al., 1982; Tapponnier et al., 1984]. On the other hand, this sense would be consistent with eastwards extrusion of the block along a southward concave fault in the early phases of the collision [Peltzer et al., 1982; Tapponnier et al., 1982]. The clockwise rotation of the southern part of the Lhasa block would also be consistent with, although smaller than the corresponding clockwise rotation of Indochina with respect to stable Eurasia (24° ± 12° [Achache et al., 1983a] or with respect to the northern part of the Lhasa block (6° ± 13°). The Indochina rotation is associated with major left-lateral strike-slip faulting along the Red River fault (Figure 18) [Tapponnier et al., 1982,1984], and one may suggest an interesting similarity, perhaps even a continuity, between the Red River and Nyainqentanglha faults [Tapponnier et al., 1982,1984; Achache et al., 1983a]. In the second hypothesis the southern part of the Lhasa block would have undergone diffuse penetrative deformation between the two right-lateral Zangbo and Nyainqentanglha-Jiali strike slip fault systems (Tapponnier et al., 1984, Figure 11D). This would imply a component of thrusting in the Nyainqentanglha, which would then appear to be a composite shear/push up oblique range (J.P. Brun, personal communication, 1984) [Tapponnier et al.,1984]. The latter interpretation agrees with the field observation that tectonic deformation is significantly more intense in the southern part of the Lhasa block. Such penetrative deformation would, in addition, account for the dispersion of magnetic declinations in the south of the Lhasa block (see previous section, Figures 9, 10, 15 and the large values of δD in Table 5).

The paleomagnetic results reported in this paper are consistent with the previous analysis of Achache et al. [1983a], where a close paleogeographic relationship between Southern Tibet and Indochina is documented. Paleolatitudes for the Upper Cretaceous agree remarkably well (12.5° N ± 5° versus 15.5° ± 9° N), and we still hold the view that the two formed the regular east-west trending active southern margin of Eurasia that was subsequently disrupted by the arrival of India.

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